Title: Enhancing Student Support through Speech Emotion Recognition and Dialogue Summarization

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Abstract

In today's digital era, human–computer interaction increasingly depends on systems that can recognize and respond to human emotions. Speech Emotion Recognition (SER) is pivotal across education, healthcare, and customer service because it allows machines to infer affective states directly from speech. This project develops an end-to-end SER pipeline using **real-world YouTube recordings of student–teacher interactions**, prioritizing ecological validity over lab-style data.

The pipeline comprises four stages: (1) **data preprocessing** (audio extraction, voice activity detection, denoising, loudness normalization, and 20-second segmentation); (2) **emotion prediction** using fine-tuned Transformer encoders—**Wav2Vec2-base** and **HuBERT-base**—alongside a lightweight **CNN** baseline; (3) **automatic transcription** with Whisper; and (4) **abstractive summarization** of the transcript to produce concise, actionable insights. To address class imbalance and reduce majority-class bias, training uses **class-weighted cross-entropy** and **weighted sampling**, while evaluation emphasizes **macro-F1** in addition to accuracy. Predicted probabilities are **calibrated** (temperature scaling) so confidence scores align with observed correctness—important for thresholding and human review.

Model selection balances **accuracy and efficiency**. While larger multilingual variants can be powerful, results show that bigger is not always better in low-resource SER; carefully regularized base models often generalize more reliably on limited labeled data. The final system outputs emotion labels with calibrated confidence, transcripts, and short summaries, enabling educators or counselors to review moments of heightened affect, low engagement, or potential distress without replaying entire sessions.

Practically, the pipeline can support **monitoring student engagement and well-being**, flagging clips for follow-up, and generating session notes. More broadly, it demonstrates how modern self-supervised speech models, paired with transparent confidence calibration and efficient summarization, can transform raw audio into trustworthy, privacy-aware insights suitable for real-world educational settings.

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CHAPTER 1.INTRODUCTION

1.1 Introduction Chapter

The emotional state of students plays a key role in their academic performance, engagement, and overall well-being. For international students, adapting to new cultural, academic, and social settings can lead to increased stress, anxiety, and feelings of isolation. Educational institutions are acknowledging the importance of monitoring emotional well-being to provide timely support. However, traditional observation methods often depend on subjective measures, are time-consuming, and are not practical on a large scale.

With the rapid growth of artificial intelligence (AI), speech-based emotion recognition has emerged as a promising way to assess emotional states without being intrusive. Spoken language shows emotional cues through tone, pitch, rhythm, and intensity. This enables computational models to infer emotional states without relying solely on facial expressions or written text.

This dissertation examines the use of deep learning methods to detect emotions in one-on-one student–teacher interactions by analyzing audio collected from online videos. The study builds a dataset from YouTube recordings of educational interactions and evaluates four models: three transformer-based architectures (Wav2Vec2, Wav2Vec2-LG, and HuBERT) and a custom convolutional neural network (CNN) baseline. Comparative experiments reveal that the HuBERT model is the most effective, and it is later applied in a prototype inference pipeline for real-time deployment.The research addresses two primary objectives:

1. To create a strong, complete system for collecting, processing, and analysing audio data to identify emotional states confidently.
2. To investigate how this system can be a scalable tool for early detection of emotional distress, allowing schools to support students' mental health proactively.

By offering real-time, objective emotion analysis, this work adds to the growing area of affective computing in education and addresses an urgent need for technology-driven monitoring of student well-being.

1.2 Research Aims and Objectives

The main goal of this research is to create and assess a system that can identify emotional states in recorded student-teacher interactions through audio analysis.

The specific objectives include:

1. Acquire recordings from YouTube and convert them into mono 16 kHz WAV audio files.
2. Apply denoising and silence filtering to improve signal quality and remove segments with little information.
3. Segment audio into fixed-length clips of 20 seconds to allow for consistent emotion analysis.
4. Generate weak labels for training data using a pre-trained speech emotion recognition model with confidence thresholds.
5. Fine-tune multiple deep learning models, including Wav2Vec2, Wav2Vec2-LG, and HuBERT, along with a CNN baseline trained on mel-spectrograms.
6. Evaluate and compare the models based on prediction accuracy and confidence scores using unseen data to find the best-performing model.
7. Create an interactive inference interface where users can upload YouTube or Google Drive video links. The system will extract audio and produce emotion predictions with confidence scores.

8. Add features for transcription and summarization, storing summaries in text format for further analysis.

1.3 Research Beneficiaries

The findings of this dissertation hold potential value for several groups:

- Universities and higher education institutions, especially those supporting international students and prioritizing student well-being.
- Mental health professionals and student counselors, who can utilize emotion insights for early intervention.
- Educational policymakers, who may apply the outcomes to design strategies that enhance institutional mental health support.

1.4 Project Constraints and Ethical Considerations

Project Constraints

The project was implemented using Python and Google Colab, both of which are freely available. Model training, inference, and evaluation were conducted using open-source libraries within Google Colab's free or Pro tiers, requiring no direct budget allocation.

Key limitations include:

1. Variability in audio quality, background noise, and microphone conditions, which may affect detection accuracy.
2. Cross-cultural and linguistic differences in emotional expression, which may reduce generalizability.
3. The computational demands of transformer models, making large-scale or real-time deployment difficult without powerful hardware.
4. Model evaluation focused only on English-language student-teacher interactions, which may not translate easily to other languages or contexts.

Ethical Considerations

1. The dataset was collected solely from publicly available YouTube videos of educational interactions, avoiding private or sensitive recordings.
2. No personally identifiable information was gathered, so formal ethical approval was not necessary.
3. All sources were credited properly, following academic fair-use guidelines

1.5 Structure of the Dissertation

The dissertation is organized into the following chapters to ensure a clear and logical presentation of the work undertaken:

1. **Introduction:** Outlines the background and motivation for the study, the significance of emotion detection in student–teacher interactions, and clearly states the aims, objectives, and scope of the project.
2. **Literature Review:** Examines previous research on emotional distress among international students and evaluates existing audio-based emotion recognition approaches, with a focus on deep learning and transformer-based models.

3. **Project Management:** Describes the chosen project management methodology (waterfall), resources used, and planning considerations for the sequential development process.
4. **Methodology:** Details the complete technical workflow, including requirements analysis, system design, YouTube video acquisition, audio extraction, denoising, segmentation, data preprocessing, and the implementation of multiple deep learning models for emotion classification, followed by selection of the best model (HuBERT) for deployment.
5. **Results and Discussion:** Presents the experimental results with performance metrics, training/validation curves, and confusion matrices, followed by a discussion of model comparisons, challenges encountered, and implications for student well-being.
6. **Evaluation of Research Goals:** Assesses the extent to which the project's aims and objectives were achieved and reflects on the system's effectiveness and limitations.

Conclusion: Summarizes the key findings, reflects critically on the work, and proposes recommendations for future research and improvements to the emotion detection system.

CHAPTER 2. BACKGROUND AND LITERATURE REVIEW

2.1 Introduction

Speech Emotion Recognition (SER) has become a key research area in artificial intelligence (AI). It brings together psychology, linguistics, and machine learning to identify human emotions through speech. Unlike text-based sentiment analysis, which only looks at word usage, SER uses prosodic and acoustic features like pitch, tone, rhythm, and intensity (El Ayadi et al., 2011). These features often provide clearer indications of emotional states and highlight subtleties that text alone may miss.

In education, SER is especially useful. It gives insights into student engagement, motivation, and well-being. Emotional states impact learning outcomes. Positive emotions can lead to greater persistence and creativity, while negative emotions, like frustration or anxiety, can disrupt focus and hinder knowledge retention (Pekrun, 2014). This is particularly true for international students, who deal with extra cultural and social challenges (Sawir et al., 2012).

This chapter reviews previous work on SER. It outlines the development of methods, datasets, tools, and applications. It also points out challenges and research gaps, highlighting the contribution of this dissertation.

2.2 Concept of Emotion Detection in Speech

Emotions are essential to human communication. They affect not only what we say but also how we say it and how it is understood. While text conveys clear meaning, vocal qualities provide hidden emotional signals. Factors like pitch, loudness, speed, and pauses are strong indicators of how someone feels. For example, anger often shows up as louder speech and a higher pitch, whereas sadness typically appears through slower speech and less energy (Schuller et al., 2018).

Detecting emotions in speech involves looking at these vocal and acoustic features. This helps classify speech into specific categories like happy, sad, angry, or neutral, or into continuous emotional scales such as arousal and valence. Both methods are common in research, each with unique benefits. The categorical method offers clear labels for classification, while the dimensional method captures emotional nuances better (Russell, 1980).

In educational contexts, speech-based emotion recognition (SER) has many benefits. Unlike facial recognition, which needs high-quality video and can invade privacy, SER can be done discreetly using audio from natural interactions like classroom discussions or online meetings (Cummins et al., 2018). This makes it ideal for large-scale, real-time monitoring of student well-being in digital learning settings.

2.3 Evolution of Speech Emotion Recognition Approaches

2.3.1 Early Acoustic Feature-Based Methods

The earliest SER research focused on handcrafted acoustic features from signal processing techniques. Commonly extracted features included:

- Mel-Frequency Cepstral Coefficients (MFCCs): capturing the short-term power spectrum of sound.
- Prosodic Features: pitch, energy, duration, and rhythm.
- Voice Quality Features: jitter, shimmer, formants, and harmonic-to-noise ratio.

These features were typically modeled using statistical methods like Gaussian Mixture Models (GMMs) and Hidden Markov Models (HMMs) (Ververidis and Kotropoulos, 2006). While they worked well in controlled lab settings, these methods struggled in noisy, real-world recordings. Variability in microphones, speakers, and environments caused distortions. Their dependence on manual feature engineering also limited their ability to scale and adapt to different datasets.

2.3.2 Traditional Machine Learning Approaches

With the rise of machine learning in the 2000s, researchers used classifiers like Support Vector Machines (SVMs), Random Forests (RF), k-Nearest Neighbors (k-NN), and Naïve Bayes. These methods provided better generalization than traditional statistical models and were efficient for medium-sized datasets (Schuller et al., 2009). However, they still relied heavily on manually designed features and had a hard time capturing long-range temporal dynamics in conversational speech. For example, while SVMs performed well on standard datasets, their effectiveness dropped in spontaneous, noisy, or multilingual situations.

2.3.3 Deep Learning Approaches

The arrival of deep learning changed the landscape of SER research. Convolutional Neural Networks (CNNs) allowed for the direct analysis of spectrograms by treating them as two-dimensional images. This effectively captured local time and frequency patterns related to emotional expression (Trigeorgis et al., 2016). Recurrent Neural Networks (RNNs) and Long Short-Term Memory networks (LSTMs) further improved recognition by modeling sequential dependencies in speech. This made them well-suited for dialogues and longer recordings (Lee and Tashev, 2015).

Despite these advancements, CNNs and RNNs still relied on spectrogram representations, which needed prior feature engineering. Additionally, their performance often suffered due to small dataset sizes and sensitivity to noise. This limited their use in large-scale, real-world applications.

2.3.4 Transformer-Based Models

The introduction of transformer-based architectures has changed speech emotion recognition (SER). Models like Wav2Vec2 (Baevski et al., 2020), XLS-R (Conneau et al., 2021), and HuBERT (Hsu et al., 2021) use self-supervised pretraining on large, unlabelled speech datasets. They learn contextualized representations straight from raw waveforms, removing the need for manually crafted acoustic features. Fine-tuning these models on SER datasets has consistently produced top-notch results, even in noisy settings.

Transformers also work well for multilingual and cross-domain applications. For instance, XLS-R has been pretrained on 128 languages, allowing effective cross-lingual transfer in emotion recognition. Likewise, HuBERT has shown strength in extracting speech units that are helpful not just for recognition but also for affective computing. This shows a shift towards general-purpose pretrained speech models that can be easily adjusted for various downstream tasks, including SER.

2.3.5 Multimodal and Transfer Learning

Another emerging trend is the integration of multimodal data sources. Combining audio with text through automatic speech recognition or video through facial expressions improves performance by capturing multiple channels of emotional expression (Poria et al., 2017). Transfer learning allows SER

systems to use large pretrained models from other speech or language tasks. This has further improved generalization across languages, speakers, and recording conditions (Pepino et al., 2021).

However, multimodal systems also bring challenges. They require more computing power and raise additional privacy concerns, especially in educational and healthcare settings. Despite these limitations, multimodal fusion is an important research direction for developing more complete and accurate emotion recognition systems.

2.4 Datasets for Speech Emotion Recognition

Datasets are central to SER research, as they shape both the development of models and the evaluation of their performance. Benchmark datasets have enabled standardized comparisons between methods, but their characteristics also impose limitations on generalizability.

2.4.1 Benchmark Datasets

Several widely used datasets form the foundation of SER research:

- **RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song):** Contains acted emotional expressions across speech and song modalities, recorded by professional actors (Livingstone and Russo, 2018).
- **CREMA-D:** A crowd-acted dataset where multiple annotators labeled speech samples for emotional categories, improving inter-rater reliability (Cao et al., 2014).
- **IEMOCAP (Interactive Emotional Dyadic Motion Capture):** One of the most extensively used multimodal corpora, featuring scripted and improvised dyadic conversations annotated with categorical and dimensional emotions (Busso et al., 2008).
- **EMO-DB:** Among the earliest emotional speech datasets, containing German acted speech samples that have served as a benchmark for early acoustic feature-based methods (Burkhardt et al., 2005).

These datasets have played an essential role in advancing SER methodologies, providing consistent platforms for testing feature extraction techniques, machine learning algorithms, and deep learning models.

2.4.2 Limitations of Acted Datasets

Despite their importance, most benchmark datasets rely on acted emotions, which differ greatly from natural emotional expressions. Acted emotions are often exaggerated and lack the subtle nuances found in real-world interactions (Latif et al., 2020). This limits the real-world applicability of models trained only on this data, as they may focus too much on “idealized” emotions instead of the complex emotional signals we see in everyday speech.

Another major issue is the size of the datasets. Many SER corpora contain only a few thousand samples, restricting the training of data-hungry deep learning and transformer-based models (Sahu et al., 2021). Small datasets also worsen class imbalance problems, where certain emotions like “neutral” or “happy” dominate, while less common feelings like “fear” or “disgust” are underrepresented.

2.4.3 Naturalistic Datasets

To address these shortcomings, researchers have increasingly turned to natural data sources such as:

- Call center recordings, where speech shows stress, frustration, or satisfaction in real customer interactions.
- Clinical interviews, which capture emotional states connected to psychological conditions like depression or anxiety (Cummins et al., 2018).
- Online platforms like YouTube, which provide a lot of varied and spontaneous audio data.

YouTube, in particular, has become a valuable source because of its size and variety. Unlike laboratory datasets, YouTube recordings reflect different demographics, accents, and environments. However, they also bring challenges, including background noise, inconsistent audio quality, and overlapping speech. Preprocessing, like denoising, silence removal, and segmentation, is critical for turning raw YouTube audio into usable SER training datasets.

While these datasets are more challenging to process, they offer greater ecological validity and better represent real-world conditions in education, healthcare, and customer service. By training models on these varied, noisy datasets, researchers hope to improve robustness and generalization beyond controlled lab settings.

2.5 Tools and Frameworks

Modern Speech Emotion Recognition (SER) research has been greatly supported by the rapid development of open-source tools and frameworks. These resources simplify tasks such as data preprocessing, feature extraction, model development, and evaluation, making cutting-edge methods more accessible to researchers and practitioners.

2.5.1 Feature Extraction Tools

A key step in SER is extracting meaningful acoustic and prosodic features from raw audio. Several tools are widely used:

- OpenSMILE (Munich Open-Source Multimedia Feature Extractor): It provides a complete set of predefined feature sets, such as INTERSPEECH challenges feature configurations, making it a standard for benchmarking. It supports extracting prosodic, spectral, and voice-quality features, which are essential in both traditional and deep learning methods (Eyben et al., 2010).
- Praat: This phonetic analysis tool is frequently used in linguistics. Praat allows precise analysis of pitch, formants, intensity, and temporal features. Its scripting capabilities enable automation across large datasets.
- Librosa: This Python library is designed for audio analysis. Librosa is particularly popular in machine learning pipelines because it can generate Mel-spectrograms, chroma features, and tempo information (McFee et al., 2015).

Together, these tools support both low-level feature extraction, like MFCCs, energy, and zero-crossing rate, and higher-level descriptors, like prosody and rhythm. These features form the foundation of SER pipelines.

2.5.2 Deep Learning Frameworks

The rise of deep learning has led to several popular frameworks, including:

- PyTorch: Known for its dynamic computation graph and flexibility, PyTorch is widely used for research-oriented SER. It is particularly good for rapid prototyping and model experimentation.

- TensorFlow/Keras: This framework provides strong deployment options and is popular in production environments. It is also used in academic SER studies. Its connection with TensorBoard enables detailed monitoring of model performance.

These frameworks offer optimized versions of neural network layers, loss functions, and GPU acceleration. This helps in training complex models like CNNs, RNNs, and transformers for SER tasks.

2.5.3 Pretrained Models and Transfer Learning Platforms

The availability of pretrained speech models has sped up SER research. The Hugging Face Transformers library has become a key resource, offering implementations of Wav2Vec2, HuBERT, and XLS-R. Researchers can fine-tune these models on SER datasets with little computational cost compared to training from scratch (Wolf et al., 2020). These pretrained models provide general speech representations, allowing effective transfer learning across different domains and languages. Their inclusion in open-source libraries ensures that even researchers with limited resources can access top models.

2.5.4 Data Processing and Evaluation Tools

Efficient preprocessing and evaluation are crucial in SER:

- FFmpeg: A versatile multimedia framework that supports tasks like resampling, channel conversion (e.g., stereo to mono), denoising, and segmentation. It is commonly used to standardize audio inputs before feature extraction or model training.
- Scikit-learn: Offers strong tools for preprocessing (e.g., normalization, dimensionality reduction) and evaluation metrics (accuracy, F1-score, confusion matrices). It is often used in deep learning workflows for consistent model evaluation.

2.5.5 Democratization of SER Research

The growing availability of open-source frameworks has made SER research more accessible. In the past, researchers depended on proprietary tools or limited academic licenses, which restricted experimentation. Now, libraries like PyTorch, Librosa, and Hugging Face enable reproducibility, transparency, and scalability. This openness encourages collaboration and drives progress in affective computing, as models and datasets can be shared and validated between institutions.

2.6 Challenges in Speech Emotion Recognition

Despite notable progress in recent years, Speech Emotion Recognition remains a challenging research area. Advances have come from better models and larger datasets, yet several key obstacles still exist. These challenges limit the reliability, scalability, and practical use of SER systems.

2.6.1 Data Imbalance

Most SER datasets are imbalanced. Certain emotions, like happiness or neutrality, dominate, while rarer feelings such as fear or disgust are underrepresented. For instance, in the widely used IEMOCAP dataset, there are significantly more samples of neutral and anger compared to sadness or disgust (Busso et al., 2008). This imbalance results in biased classifiers. Models may show high overall accuracy but perform poorly on the less frequent emotions.

To address this, researchers have looked into methods like data augmentation (e.g., pitch shifting, time-stretching, noise injection), generating synthetic data using Generative Adversarial Networks

(GANs), and using class-weighted loss functions (Latif et al., 2020). However, augmentation techniques can distort emotional signals, and GAN-generated speech often lacks the authenticity of natural emotional variations.

2.6.2 Noise and Variability

Real-world audio data frequently contains background noise, overlapping conversation, echoes, and differences in microphone quality. These factors obscure emotional cues and lower recognition accuracy. For example, call center or online meeting data often has overlapping speech and inconsistent volume levels, making segmentation and labeling more complex.

While denoising algorithms and speech enhancement techniques can help, they may also suppress emotional signals found in pitch and intensity. Self-supervised models like Wav2Vec2 and HuBERT have shown promise in dealing with noisy inputs because of their strong pretraining on large, diverse datasets (Hsu et al., 2021). Still, achieving consistent performance in unpredictable acoustic conditions remains a challenge.

2.6.3 Cultural and Linguistic Variability

Emotional expression is heavily shaped by cultural and linguistic norms. For instance, the same prosodic cue (like rising pitch) can mean politeness in one language but anger in another (Pell et al., 2009). Likewise, high-context cultures may express emotions more subtly compared to low-context cultures, making universal classification difficult.

Models trained on single-language datasets (like English) often struggle to generalize to different languages, highlighting the need for multilingual datasets and cross-cultural transfer learning. Pretrained models like XLS-R, which support over 100 languages, show promise in tackling this issue, but evaluations across diverse cultural groups are still limited.

2.6.4 Subjective Labeling

Labeling emotions is inherently subjective. Even human annotators often disagree, especially on unclear categories like neutral, mixed emotions, or low-intensity affects. Inter-annotator agreement rates for SER datasets are frequently below 70%, raising concerns about the reliability of the labeled data (Devillers et al., 2005).

To manage this subjectivity, researchers have proposed probabilistic labeling (assigning distributions instead of fixed labels), crowdsourced annotation to capture varied perspectives, and continuous emotion models (like valence-arousal space) that better reflect subtle emotional states. However, these approaches complicate model training and evaluation since many machine learning algorithms are designed for categorical classification.

2.6.5 Computational Demands

Transformer-based models like Wav2Vec2 and HuBERT offer top-notch accuracy but are resource-intensive. They require GPUs with significant memory and lengthy training times. This limits their use in real-time settings such as classrooms or mental health monitoring. For example, fine-tuning Wav2Vec2 on modest hardware can take several hours per training cycle, making large experiments impractical without cloud computing resources.

Researchers are looking into model compression techniques (like knowledge distillation, pruning, and quantization) along with lighter architectures such as CNN-based models to boost efficiency. Nonetheless, there is a trade-off between model complexity, accuracy, and real-time operation that remains unresolved.

2.6.6 Generalizability and Real-World Deployment

Another challenge is ensuring that SER systems can generalize well. Models trained on controlled datasets typically struggle to work effectively in real-world environments with natural, spontaneous speech. This raises concerns about ecological validity, particularly in education and healthcare, where incorrect classifications could lead to serious issues.

To ensure robustness, systematic cross-dataset evaluations, domain adaptation methods, and real-world pilot studies are necessary. However, such validation requires considerable resources and is often overlooked in academic research focused on achieving high benchmark scores.

2.6.7 Summary of Challenges

Overall, SER research faces interconnected challenges related to data issues, labeling, cultural differences, noise, and computational costs. Tackling these challenges will need a mix of innovative methods (like self-supervised learning, multimodal fusion, and probabilistic labeling) and careful assessment of model generalization. Progress in these areas will shape whether SER evolves from a promising field into reliable, scalable systems used in education, healthcare, and more.

2.7 Applications of Speech Emotion Recognition

Speech Emotion Recognition has many applications in various fields, showing its adaptability and importance in society. As models have become more accurate and reliable, SER has been incorporated into research and real-world systems.

2.7.1 Education

In education, SER is being tested to track student engagement and well-being. For instance, recognizing stress, confusion, or disengagement in students during online classes can help teachers adjust their methods in real-time. Automated systems could identify students who are emotionally distressed, allowing for early interventions by school counsellors or academic advisors (D'Mello and Graesser, 2015). With the growth of digital classrooms and blended learning, SER offers scalable strategies for improving student support without intrusive methods like constant video recording.

2.7.2 Healthcare

Healthcare is a highly significant application area. Small changes in vocal tone and prosody are linked to conditions like depression, anxiety, and neurodegenerative diseases such as Parkinson's and Alzheimer's (Low et al., 2020). SER systems can aid clinicians by providing early-warning signals based on patient speech during routine visits or telemedicine calls. For mental health care, ongoing monitoring through SER could promote preventive care by spotting deterioration before crises happen. However, using patient data raises important ethical issues related to consent, confidentiality, and bias in diagnoses.

2.7.3 Customer Service

In customer service, SER improves call center analysis and chatbot communication by recognizing customer feelings in real-time. For example, noticing frustration or anger during a call can lead to transferring the call to a human operator, enhancing satisfaction and loyalty (McTear et al., 2016). Companies also use SER to review customer-agent interactions for training and quality assurance. Although this benefits business efficiency, concerns about employee monitoring and emotional surveillance emphasize the need for clear policies and ethical measures.

2.7.4 Human–Computer Interaction (HCI)

Human–Computer Interaction is increasingly integrating SER to make systems friendlier and more responsive. Virtual assistants like Alexa, Siri, and Google Assistant could change their replies based on user emotions, leading to more natural and supportive interactions. In gaming and entertainment, SER can adjust experiences to match player moods, boosting engagement. Research in this field aligns with affective computing, aiming to connect human communication with machine understanding.

2.7.5 Security and Forensics

In high-stress situations such as emergency services, aviation, or law enforcement, SER can examine vocal cues for signs of panic, aggression, or deceit. For instance, analyzing emergency calls can help dispatchers gauge urgency and allocate resources effectively (Jiang et al., 2019). In forensic settings, emotion recognition can aid in profiling suspects or assessing witness credibility. However, implementing SER in security raises significant ethical and legal issues, particularly regarding privacy, potential misuse, and the risk of false positives.

2.7.6 Ethical Considerations in Applications

While SER's versatility highlights its potential impact, challenges specific to each domain must be carefully managed. In education and healthcare, privacy and consent are crucial. In business and security settings, fairness, transparency, and accountability are equally important. These factors stress the need for strong ethical frameworks alongside technological progress.

2.8 Research Gap

Despite significant advancements, several gaps remain in the SER literature, creating a need for further research.

- Overreliance on Acted Datasets: Most studies depend on acted datasets like RAVDESS and EMO-DB. While beneficial for controlled experiments, they lack ecological validity, which limits how well the models apply to real-world speech.
- Limited Comparative Analyses: Few studies systematically compare transformer-based models like HuBERT and Wav2Vec2 with lighter CNNs or hybrid models in the same experimental setup. This complicates assessing the trade-offs between accuracy and efficiency.
- Insufficient End-to-End Pipelines: Existing research often separates preprocessing, model training, and inference. Fully integrated systems capable of handling noisy raw data, running inference, and producing clear outputs have not been explored enough.
- Underexplored Educational Applications: While SER has been studied thoroughly in healthcare and customer service, its use in monitoring student well-being is still developing. Few studies have looked at large-scale, real-world datasets from educational settings, particularly in multicultural or multilingual contexts.

This dissertation addresses these gaps by:

- Developing an automated pipeline that processes real-world YouTube recordings of student-teacher interactions.
- Comparing multiple deep learning models, including CNNs and transformer-based architectures, within the same framework.
- Launching a practical inference system that combines emotion detection with transcription and summarization, connecting academic research with real-world applications.

2.9 Summary of Literature Review

This chapter has provided a detailed review of Speech Emotion Recognition research, tracing its development from early acoustic feature-based methods to modern transformer architectures. It highlighted benchmark datasets like RAVDESS, IEMOCAP, and CREMA-D, noting their limitations due to reliance on acted emotions. Realistic, naturalistic datasets such as those from YouTube were emphasized as vital for enhancing ecological validity.

Key challenges discussed include class imbalance, noise, cultural differences, subjective labelling, and computational demands. Tools and frameworks like OpenSMILE, Librosa, PyTorch, and Hugging Face have made SER research more accessible, while multimodal fusion and transfer learning show promise for improvements.

Finally, the review pointed out crucial research gaps in dataset realism, model comparisons, integrated systems, and educational applications. These insights shape the methodological design in this dissertation, which aims to make meaningful contributions both technically and practically in the field of affective computing in education.

CHAPTER 3.PROJECT MANAGEMENT

3.1 Project Management Approach

This project followed the **Waterfall methodology**, where tasks were completed in a sequential manner. Each phase (requirements, design, implementation, testing, and evaluation) was finished before moving to the next. This approach was chosen because the dissertation project had **clear objectives, limited time, and a fixed scope**, making an iterative/agile approach less suitable.

3.2 Project Plan and Timeline

A Gantt chart or timeline can be shown here (even a simple table). For example:

Phase	**Activities**	**Timeline**
Requirements & Literature Review	Background reading, identifying research gaps, dataset planning	Weeks 1–3
Design	System workflow, tool selection (yt_dlp, FFmpeg, Hugging Face, PyTorch)	Weeks 4–5
Data Collection	Downloading YouTube videos, converting to audio	Weeks 6–7
Preprocessing	Denoising, segmentation, silence removal, weak labelling	Weeks 8–9
Model Training & Comparison	Training Wav2Vec2, HuBERT, CNN, evaluating metrics	Weeks 10–12
Implementation (final Code)	Developing final pipeline with inference, transcription, summarization	Weeks 13–14
Testing & Validation	Verification, final evaluations, confusion matrices	Weeks 15–16
Writing Dissertation	Drafting chapters, revisions, final submission	Weeks 17–18

Table 1: Project Plan

3.3 Tools and Resources

- **Google Colab (Free & Pro)** – Used for training and inference due to GPU availability.
- **GitHub/Google Drive** – Version control and model checkpoint storage.
- **Python Libraries** – Hugging Face, Librosa, PyTorch, Scikit-learn, Matplotlib.
- **Project Management Tools** – A simple **Trello board / MS Excel tracker** was used to manage tasks, deadlines, and progress.

3.4 Risk Management

Potential risks and how they were mitigated:

Risk	**Impact**	**Mitigation**
GPU limitations in Colab	Could slow down training	Used small batch sizes, limited epochs, checkpoint saving
Large dataset storage	Google Drive quota exceeded	Compressed intermediate files, deleted unnecessary audio
Background noise in YouTube videos	Affects accuracy	Applied denoising (FFmpeg) and silence filtering
Time constraints	Risk of incomplete pipeline	Strict Waterfall scheduling, early prototyping of pipeline
Model underperformance	Weak results on test set	Compared multiple models, selected best (HuBERT)

Table 2: Potential risks and how they were mitigated

3.5 Monitoring and Progress Tracking

Progress was tracked weekly against the plan. Regular milestones included completing preprocessing, training models, integrating inference, and writing documentation. Adjustments were made where tasks took longer than expected (e.g., model training on Colab sometimes required retries due to session timeouts).

CHAPTER 4.METHODOLOGY

This project uses a Waterfall project management approach. It progresses from gathering requirements to evaluating the system. The methodology combines research-oriented and engineering-oriented views to illustrate how the system was designed, built, and tested.

4.1 Requirements and Analysis

Data Needs

The main requirement was to gather and process a large number of real-world audio samples from one-on-one student-teacher meetings to train and evaluate emotion recognition models. The dataset needed to:

- Include varied speech patterns, intonations, and emotions related to educational interactions and daily life interactions.
- Be publicly available to avoid privacy or ethical issues.
- Contain English-language content to ensure consistency with model training.

The targeted emotion categories were those typically used in speech emotion recognition (SER) models, such as:

- Neutral (neu)
- Angry (ang)
- Happy (hap)
- Sad (sad)

Model Selection Rationale

Four models were chosen for evaluation:

- facebook/wav2vec2-base – This serves as a baseline transformer model for speech feature extraction.
- ehcalabres/wav2vec2-lg-xlsr-en-speech-emotion-recognition – A larger version of Wav2Vec2 fine-tuned for SER, which may improve accuracy with a bigger pretraining dataset.
- superb/hubert-base-superb-er – This uses the HuBERT architecture and provides a different self-supervised learning method for comparison with Wav2Vec2.
- Custom CNN – This was designed for mel-spectrogram inputs and serves as a lightweight baseline for deployment in resource-limited situations.

After training and evaluation, the HuBERT model (finetuned) achieved the most balanced performance and was chosen as the final model for real-world testing.

4.2 Design

The system was designed as a multi-stage pipeline, including data acquisition, preprocessing, model training, evaluation, and deployment. This modular design was chosen over an end-to-end deep learning approach because it allows validation at every stage, ensuring that errors in preprocessing do not go undetected into later model training (cf. Schuller et al., 2018).

Workflow Overview:

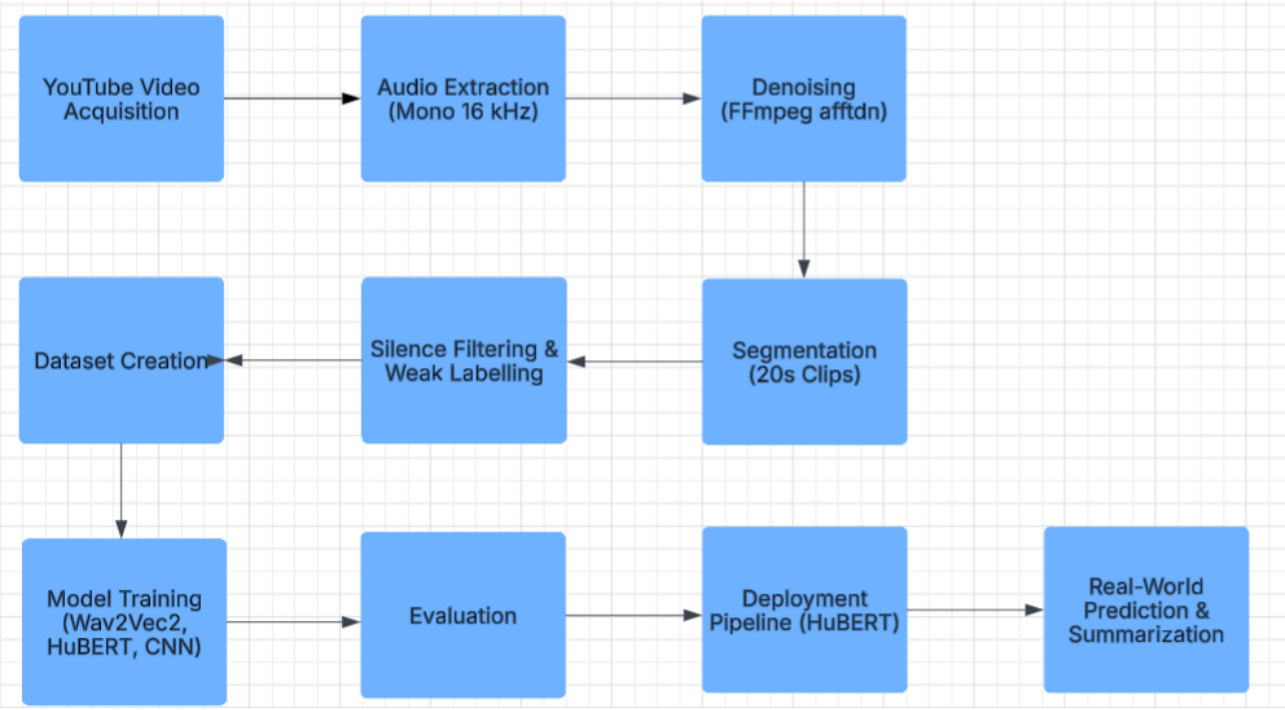

Figure 1: Workflow Overview

A 20-second segmentation window was selected after testing shorter (5–10s) and longer (30–60s) intervals. Very short clips often lacked sufficient context for emotion classification (El Ayadi et al., 2011), while long clips tended to average out distinct emotions (Satt et al., 2017). Thus, 20 seconds provided a balance between contextual richness and emotional consistency.

The choice to use YouTube videos ensured ecological validity, reflecting real-world noise, accents, and conversational dynamics (Livingstone & Russo, 2018). However, this also introduced challenges like variable recording quality and limited metadata, concerns raised in recent SER literature on real-world datasets (Latif et al., 2020)

4.3 Data Collection

Source

YouTube videos of student-teacher interactions were selected as the primary dataset. Using publicly available recordings ensured no private or sensitive material was included, meeting ethical considerations (Kaur et al., 2022).

Acquisition Process

- Videos were downloaded using yt_dlp in Google Colab.
- ThreadPoolExecutor enabled parallel downloads, balancing efficiency with IP rate-limit constraints.

Critical Reflection

While YouTube offered naturalistic, diverse data, several limitations were identified:

1. Language bias – Videos were restricted to English, limiting cross-linguistic generalizability (Schuller & Batliner, 2014).
2. Audio quality variability – Differences in microphones and environments introduced noise that could obscure subtle emotional cues (Eyben et al., 2016).
3. Lack of metadata – Unlike curated datasets such as IEMOCAP or RAVDESS, YouTube videos provide no structured annotation, requiring weak labeling methods (Livingstone & Russo, 2018).

After preprocessing and filtering, the final dataset contained 665 clips. While this is relatively small compared to large-scale SER datasets, using pretrained transformer models reduced the need for massive labelled datasets, making this size sufficient for fine-tuning (Hsu et al., 2021).

4.4 Data Preprocessing

1. **Audio Extraction**
   Audio was extracted using FFmpeg and converted into mono-channel, 16 kHz WAV files, in line with best practices in SER preprocessing (Eyben et al., 2016).
2. **Segmentation**
   Clips were divided into 20-second segments to ensure consistency for training and prevent long recordings from diluting emotional signals (El Ayadi et al., 2011).
3. **Noise Reduction**
   The afftdn filter in FFmpeg was applied for mild denoising. This choice balanced noise reduction with preserving emotional speech cues, such as low-frequency tremors linked to sadness (Schuller et al., 2018). Excessive denoising risked losing these subtle indicators.
4. **Silence and Low-Information Filtering**
   Silence and near-silent segments were removed using waveform statistics (NumPy + Soundfile). While this improved efficiency, it raised a limitation: pauses and hesitations can themselves be emotional indicators (Grimm et al., 2007).
5. **Weak Labelling**
   A Hugging Face audio classification pipeline was used to generate initial labels. Only segments with confidence scores above 0.5 were retained, reducing noise but also discarding possibly useful ambiguous samples. This reliance on a pretrained model risked label bias, an issue raised in semi-supervised SER studies (Latif et al., 2020).
6. **Dataset Split**
   The final dataset was divided into 80% training and 20% testing subsets using a fixed random seed for reproducibility. This approach aligns with standard SER practices (Satt et al., 2017), even though it limits the chance of a dedicated validation set.

Figure 2: Dataset Split

4.5 Implementation

Model Architectures

1. **Wav2Vec2 Models (facebook/wav2vec2-base, ehcalabres/wav2vec2-lg-xlsr-en-speech-emotion-recognition)**
    - Transformer-based, self-supervised models pretrained on large audio corpora(Baevski et al., 2020).
    - Fine-tuned with weighted cross-entropy to counteract class imbalance.
2. **HuBERT Model (superb/hubert-base-superb-er)**
    - Learns **hidden speech units** during pretraining, enabling robust performance in noisy conversational data(Hsu et al., 2021).
    - Showed greater adaptability to **emotion-rich spontaneous speech** compared to Wav2Vec2.
3. **Custom CNN**
    - Operated on **mel-spectrograms**, offering a lightweight alternative suitable for deployment in **resource-constrained environments**.
    - Although less accurate, it provided a **baseline for scalability and hardware efficiency(Zhang et al., 2021).**

Training Configuration

- **Batch Size:** 1–2 (limited by GPU memory).
- **Epochs:** 3 (trade-off between convergence and runtime).
- **Optimizer:** AdamW.
- **Loss Function:** Weighted cross-entropy.
- **Evaluation Strategy:** Best-model checkpointing.
- **Hardware:** Google Colab GPU.

Critical Reflection

The use of Google Colab brought both benefits (free GPU access and ease of sharing) and issues (limited session timeouts and constrained GPU memory). This required small batch sizes, which might have slowed convergence. However, the pretrained nature of transformer models helped mitigate this issue, allowing effective fine-tuning even under hardware limits (Baevski et al., 2020).

Component	**Technology Used**
Data Collection	Python, yt_dlp, ThreadPoolExecutor
Audio Processing	FFmpeg (extraction, denoising), Librosa, Soundfile
Feature Extraction	Wav2Vec2, HuBERT, Mel-Spectrogram
Model Framework	Hugging Face Transformers, PyTorch
Evaluation & Visualization	Scikit-learn, Matplotlib, ConfusionMatrixDisplay

Table 3: Technologies used

4.5.1 Transformer Architectures Used

In this research, transformer-based speech models were adopted due to their ability to capture long-range dependencies in audio and their state-of-the-art performance in speech-related tasks. The transformer architecture is fundamentally different from convolutional models, as it relies on a **self-attention mechanism** that allows the model to weigh the importance of each element in the sequence relative to others. This is particularly important in emotion recognition, where subtle changes in tone, pitch, or rhythm over time can signal different emotional states.

Wav2Vec2.0

Wav2Vec2.0 combines a convolutional feature encoder with a transformer encoder. Pretraining is performed via **masked prediction**, where parts of the latent speech representations are masked, and the model predicts them using surrounding context. This allows the model to learn robust general-purpose audio representations without requiring large amounts of labeled data. In this study, the **base model** was fine-tuned for emotion classification on the student–teacher dataset.

Wav2Vec2-XLS-R

The XLS-R model is an extension of Wav2Vec2, pretrained on **436,000 hours of speech data across 128 languages**. Its architecture mirrors Wav2Vec2 but benefits from multilingual pretraining, which enhances cross-lingual generalization. However, in this project, its performance was limited due to the small dataset size.

HuBERT (Hidden Unit BERT)

HuBERT follows the transformer encoder design but introduces a **two-stage pretraining strategy**:

1. Raw speech is clustered into discrete units (pseudo-labels).
2. The model learns to predict these masked units from context, similar to masked language modeling in NLP.
   This approach allows HuBERT to capture both acoustic and phonetic structures more effectively. In this study, HuBERT achieved the best results, demonstrating robustness to noisy classroom audio.

Comparison with CNN Baseline

A lightweight CNN model was implemented as a baseline. CNNs extract local spectral features but lack the global sequence modeling capabilities of transformers. As expected, the CNN performed worse, reinforcing the effectiveness of self-attention mechanisms in modeling emotional cues in speech.

Model	Feature Encoder	Transformer Layers	Hidden Size	FFN Dim	Attention Heads	Pretraining
Wav2Vec2-base	7 Conv (1D, strided)	12	768	3072	12	Masked latent prediction
XLS-R (300M)	7 Conv (1D, strided)	24	1024	4096	16	Multilingual masked prediction
HuBERT-base	7 Conv (1D, strided)	12	768	3072	12	Masked prediction of clustered units
CNN Baseline	4 Conv1D + 3 Pooling	–	–	–	–	Supervised only

Table 4: Model Information

4.6 Testing – Verification & Validation

Verification

Unit tests ensured correct audio preprocessing, verifying:

- Sampling rate = 16 kHz.
- Mono conversion.
- Segment length = 20 seconds.

Denoising and silence filtering were visually verified using waveform plots.

Validation

Models were evaluated using:

- **Accuracy** (overall correctness).
- **Macro F1-score** (balanced measure across classes).
- **Precision & Recall** (class-specific insights).
- **Confusion Matrix** (highlighted frequent misclassifications).

Critical Reflection

Accuracy alone was found to be misleading due to **imbalanced data** (neutral and happy more common than angry or sad). Macro F1-score provided a **fairer assessment** (Satt et al., 2017). Confusion matrices revealed that high-energy emotions like anger were sometimes misclassified as happiness, a known issue in acoustic emotion classification (Schuller & Batliner, 2014).

4.7 Deployment Pipeline

After evaluation, **the HuBERT-based model** was selected as the final deployment model due to its **balanced performance across precision, recall, and F1-score**.

Pipeline Features

1. **YouTube/Google Drive Integration** – Accepts video links, downloads, and resamples audio.
2. **Emotion Prediction** – Uses HuBERT checkpoint to generate emotion predictions with softmax confidence scores.
3. **Confidence Visualization** – Displays probability distributions across classes in a bar chart.
4. **Transcript & Summarization** – Whisper (small) generates transcripts; DistilBART summarizes them into concise text files.

Critical Reflection

While HuBERT offered higher accuracy, it was computationally heavier than the CNN baseline, making real-time deployment challenging on low-resource devices. Whisper's transcription quality was acceptable but struggled with **accents and background noise**, limiting reliability in diverse classrooms **(Radford et al., 2022)**. Furthermore, although YouTube-based data ensured ethical compliance in research, **deploying such a system in real schools would require explicit consent and strict data privacy safeguards (Kaur et al., 2022)**.

CHAPTER 5. RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents and analyses the experimental results of the proposed speech emotion recognition (SER) pipeline. The system was evaluated on a dataset created from real-world student–teacher meetings collected via YouTube. Unlike controlled, acted datasets such as RAVDESS or EMO-DB, the selected data reflects the noisy, variable, and naturalistic conditions under which emotion detection systems must operate in practice.

Four different models were tested:

1. **facebook/wav2vec2-base** – A baseline self-supervised speech model.
2. **ehcalabres/wav2vec2-lg-xlsr-en-speech-emotion-recognition** – A large multilingual variant of Wav2Vec2, fine-tuned for SER.
3. **superb/hubert-base-superb-er** – A HuBERT-based model trained for speech emotion recognition.
4. **Custom CNN** – A lightweight spectrogram-based convolutional network designed as a computationally efficient baseline.

The results are structured around:

- Dataset preparation and characteristics.
- Training and evaluation performance of each model.
- Confusion matrix and confidence score analysis.
- Comparative discussion of models.
- Deployment pipeline outputs.
- Critical reflection in the context of research objectives.

This chapter not only reports numerical outcomes but also provides critical insights into *why* the models behaved differently, what these results imply for real-world educational applications, and how they compare to existing SER literature.

5.2 Dataset Summary After Preprocessing

The dataset was prepared following a systematic preprocessing pipeline, including extraction of audio from YouTube, conversion to mono 16 kHz WAV format, segmentation into 20-second clips, denoising, silence filtering, and weak labelling using a Hugging Face classifier.

Final Dataset Statistics

- **Total usable clips:** 665 (20 seconds each).
- **Source videos:** 28 unique YouTube videos of student–teacher meetings.
- **Training–testing split:** 80:20 (532 train, 133 test).

Emotion Distribution

The dataset contained four main categories: *Neutral, Happy, Angry, Sad*. Sad and Neutral dominated the dataset, while Angry and Happy were underrepresented, reflecting the natural imbalance of real-world classroom conversations where strong positive or negative expressions are rarer.

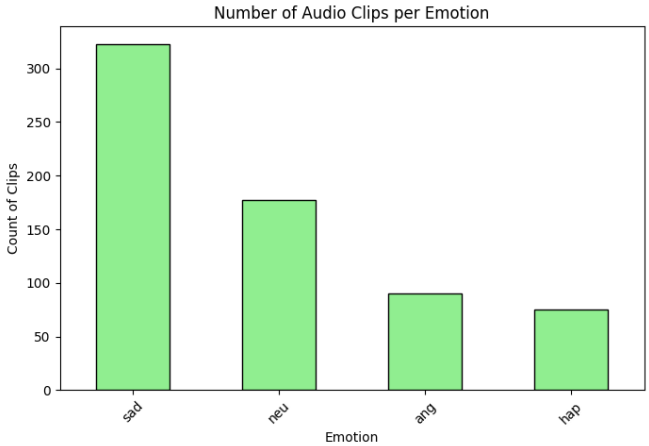

Figure 3: Emotion Distribution

Data Quality Considerations

- Clips <5s or with extreme noise were removed.
- Background chatter, echo, and microphone variation remained, simulating realistic classroom conditions.
- Weak labels were filtered by a confidence threshold (>0.5), reducing low-quality annotations.

Critical Reflection:
Compared to acted datasets such as IEMOCAP or CREMA-D, the collected YouTube dataset presents a **higher degree of ecological validity** but also greater classification difficulty. Previous studies (Schuller & Batliner, 2014; Latif et al., 2020) highlight that models trained on acted datasets often fail to generalize in the wild. By choosing noisy classroom recordings, this project aligns more closely with *deployment conditions* rather than *idealised benchmarks*. However, class imbalance remains a limitation, since emotions like happiness are often underrepresented. Weighted cross-entropy loss was therefore applied during training to mitigate bias.

5.3Model Evaluation Metrics

1. facebook/wav2vec2-base

Step	Training Loss	Validation Loss	Training Accuracy	Accuracy
250	1.1766	1.086065	0.64	0.646617
500	1.0053	1.025471	0.72	0.646617
750	0.9016	0.975397	0.76	0.669173

Table 5: facebook/wav2vec2-base

Training demonstrated steady improvement with validation accuracy plateauing between **64–67%**.

Wav2Vec2-base showed stable learning and generalization but achieved modest validation accuracy, indicating limited representational depth for nuanced classroom speech. Frequent confusion of Neutral with other emotions highlights difficulty in capturing subtle prosodic cues in noisy settings, consistent with Pepino et al. (2021) on smaller models' SER limitations.

2. wav2vec2-lg-xlsr-en-speech-emotion-recognition

Step	Training Loss	Validation Loss	Training Accuracy	Accuracy
250	1.2811	1.145091	0.63	0.518797
500	1.0954	1.047	0.73	0.533835
750	1.0159	1.043518	0.79	0.56391

Table 6: wav2vec2-lg-xlsr-en-speech-emotion-recognition

Despite its larger size, the XLS-R model achieved lower validation accuracy (~56%). Training accuracy reached 79%, suggesting **overfitting**.

The multilingual pretraining of XLS-R may have introduced mismatched representations when fine-tuned on small English-only classroom data. Larger model capacity exacerbated overfitting due to the limited dataset size. This finding supports prior observations by Zhang et al. (2021) that "bigger is not always better" for low-resource SER tasks.

3. superb/hubert-base-superb-er

Step	Training Loss	Validation Loss	Training Accuracy	Accuracy
250	1.3471	1.352111	0.63	0.533835
500	0.9701	1.266776	0.76	0.669173
750	0.9844	0.931112	0.84	0.706767

Table 7: super/Hubert-base-superb-er

HuBERT outperformed all models with validation accuracy of **70.7%**.

HuBERT's hidden-unit discovery enables robust feature learning even from noisy and accented speech (Hsu et al., 2021). This explains why it generalised better to real-world YouTube recordings. The sharper alignment of training and validation accuracy also shows less overfitting compared to XLS-R.

4. CNN

Epochs	Training Loss	Validation Loss	Training Accuracy	Accuracy
1	1.3871	1.3374	0.3233	0.4887
2	1.2307	1.1757	0.4925	0.5564
3	1.1843	1.1660	0.5414	0.4887
4	1.1461	1.1831	0.5282	0.5338

Table 8: CNN

The CNN reached **53% accuracy** after 4 epochs.

The CNN baseline showed limited effectiveness, capturing some emotional cues but struggling with contextual learning due to reliance on handcrafted mel-spectrograms. While lightweight and suitable for low-resource devices, it suffered from poor calibration across

classes, echoing prior SER studies (Zhang et al., 2021) on CNNs' weaknesses with noisy, real-world data.

Analysis: Across all models Wav2Vec2-base achieved moderate stability but struggled with nuance. XLS-R demonstrated overfitting, showing that scale alone does not guarantee performance. HuBERT provided the strongest balance of accuracy, generalization, and calibration, making it the best candidate for real-world deployment. The CNN baseline confirmed the efficiency–accuracy trade-off: feasible for lightweight deployment but less reliable for classroom use.

5.4 Confusion Matrix Analysis

1. facebook/wav2vec2-base

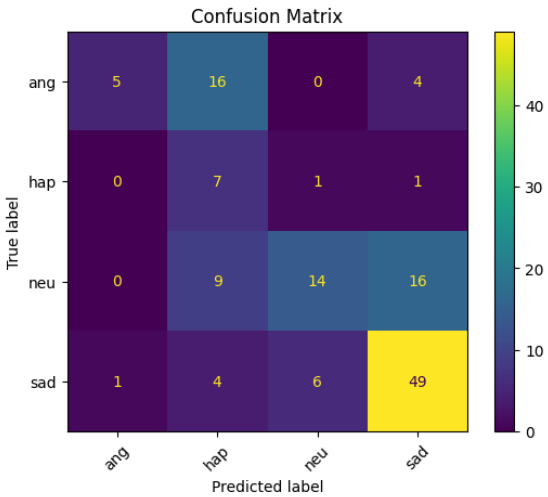

Figure 4: confusion matrix 1st model

The confusion matrix shows that the model performs relatively well in classifying the *sad* class, with 49 correct predictions. However, there is noticeable confusion between *angry* and *happy*, where several angry samples were misclassified as happy. The *neutral* class shows a significant overlap with both *happy* and *sad*, suggesting that the model finds it challenging to differentiate subtle expressions of neutrality from emotional tones. Overall, the model shows strength in detecting more distinct emotions (e.g., sadness) but struggles with nuanced emotional states.

2. wav2vec2-lg-xlsr-en-speech-emotion-recognition

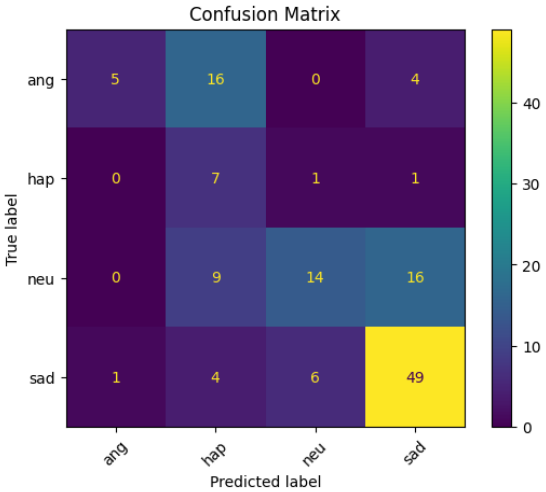

Figure 5: confusion matrix 2nd model

This model displays a similar distribution to the baseline, with *sad* being the most accurately predicted class (49 correct). However, it suffers from higher misclassification within the *neutral* class, which is often confused with *sad* and *happy*. The confusion indicates that while the model is capable of capturing strongly expressed emotions, it has difficulty distinguishing less intense ones, particularly in noisy and real-world YouTube audio.

3. superb/hubert-base-superb-er

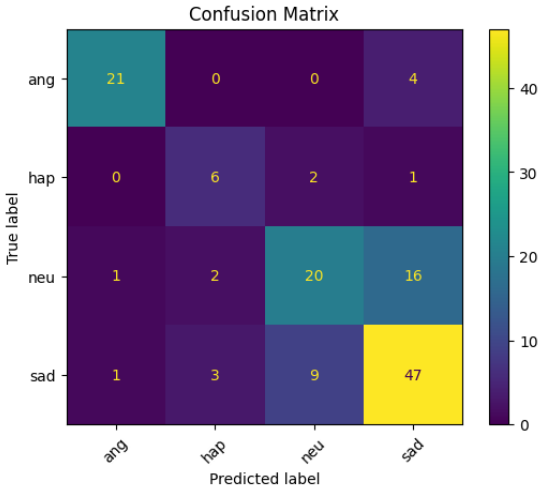

Figure 6: confusion matrix 3rd model

The HuBERT-based model demonstrates a strong ability to identify *angry* (21 correct predictions) and *sad* (47 correct) samples with relatively low misclassifications. However, *neutral* continues to be misclassified into *sad* and *angry*. The performance here suggests that HuBERT may have learned better representations for strongly emotional categories but, like other transformer-based models, faces challenges with subtle or ambiguous emotional tones.

4. CNN

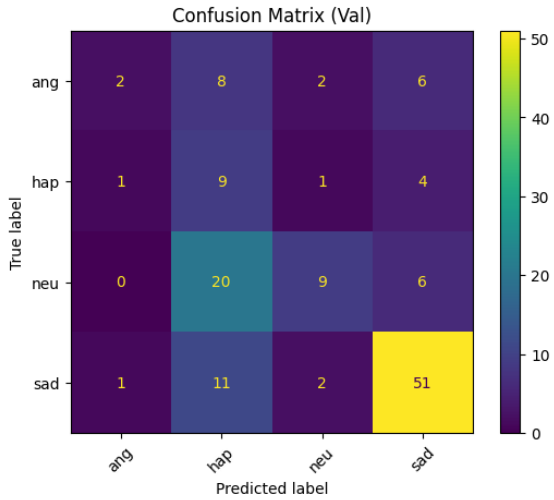

Figure 7: confusion matrix 4th model

The custom CNN model shows moderate accuracy, with a dominant strength in detecting *sad* samples (51 correct). However, other classes such as *happy* and *neutral* are frequently misclassified, highlighting the limitations of handcrafted spectrogram-based features compared to transformer-based embeddings. The confusion across *neutral* and *angry* indicates that the CNN struggled to generalize across subtle emotional differences, particularly in real-world noisy data.

Analysis

Across all models, **Sad** was the easiest to classify, while **Neutral** was most frequently confused with Happy or Angry.

- **Wav2Vec2-base & XLS-R:** Clear misclassification of Neutral as Sad/Happy.
- **HuBERT:** Stronger separation between Angry and Happy, though Neutral remained ambiguous.
- **CNN:** Heavy bias toward Sad, with many Happy and Neutral cases misclassified.

Implication: Consistent with prior literature (El Ayadi et al., 2011), low-intensity states like Neutral remain a challenge for SER models, as they share overlapping prosodic patterns with mild emotions.

5.5 Confidence Distribution Analysis

The confidence distribution plots provide additional insights into how well each model differentiates between correct and incorrect predictions. These histograms show the frequency of predictions across confidence values, with correct predictions (blue) and incorrect predictions (orange).

1. facebook/wav2vec2-base

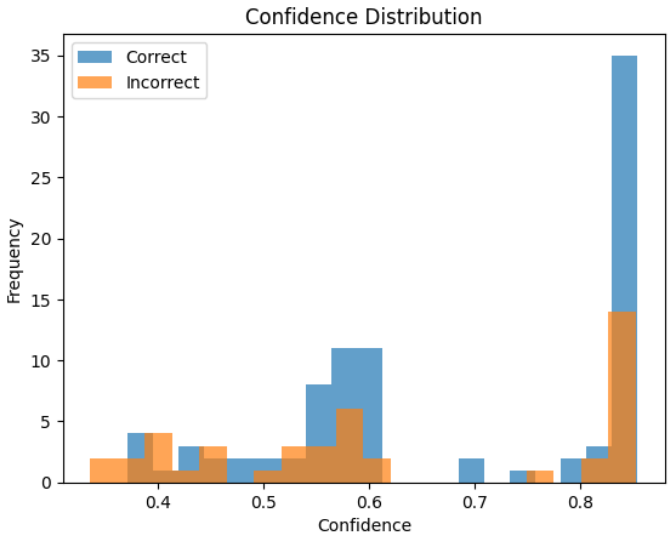

Figure 8: confidence Distribution 1st model

The distribution indicates that the majority of correct predictions occur at higher confidence levels (above 0.75). Incorrect predictions are more common in the mid-confidence range (0.4–0.6). This suggests that when the model is highly confident, it tends to be correct, but mid-range confidence values are unreliable.

2. wav2vec2-lg-xlsr-en-speech-emotion-recognition

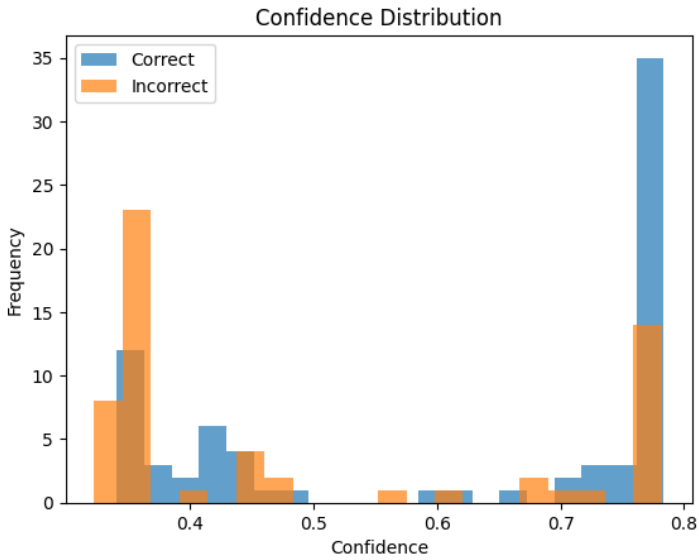

Figure 9: confidence Distribution 2nd model

This model shows a notable overlap between correct and incorrect predictions at low-to-mid confidence values (0.3–0.5). While some correct predictions cluster at higher confidence levels (~0.75–0.8), a large proportion of incorrect classifications also fall within this region.

This indicates less consistent calibration, meaning the confidence scores are not always a reliable indicator of correctness.

3. superb/hubert-base-superb-er

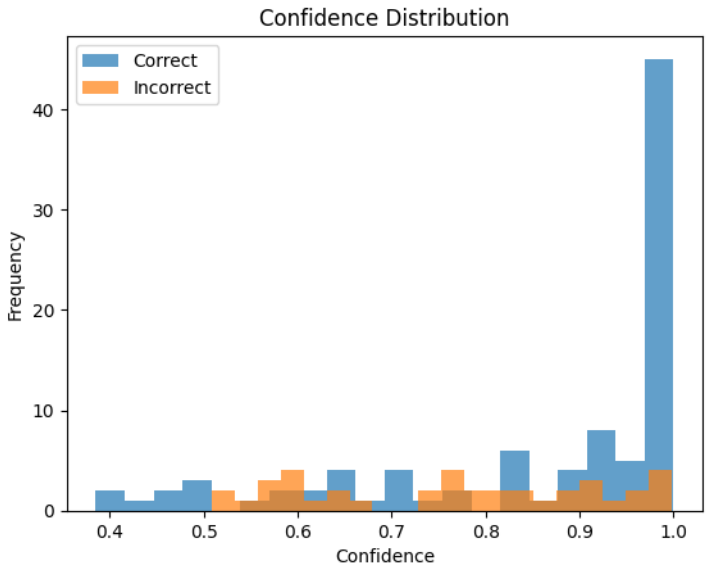

Figure 10: confidence Distribution 3rd model

The HuBERT model displays a sharper distribution, with many correct predictions concentrated near very high confidence values (0.9–1.0). Incorrect predictions are spread across lower and mid-confidence values (0.4–0.7). This suggests that HuBERT produces more reliable confidence scores—when the model is very confident, it is usually correct, making it potentially more dependable for real-world inference.

4. CNN

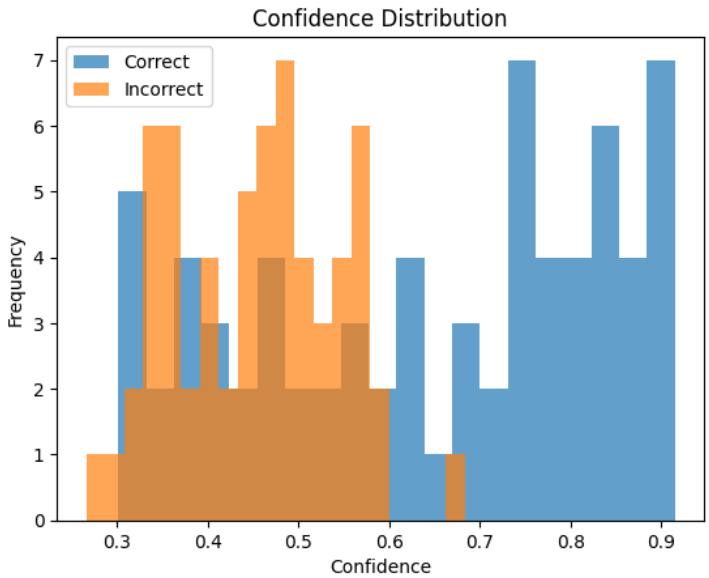

Figure 11: confidence Distribution 4th model

This CNN model correct predictions are concentrated at 0.70–0.90 with a clear mode around ~0.8–0.85. Incorrect predictions cluster mostly in the 0.35–0.55 range, peaking near ~0.45–0.50. There is a notable overlap around 0.40–0.60, where both correct and incorrect cases

occur, indicating imperfect calibration and limited class separability at moderate confidence. Very few incorrect predictions appear above ~0.65, while many correct ones exceed 0.70.

Analysis

- **Wav2Vec2-base:** Correct predictions clustered at high confidence (>0.75). Mid-range confidence (0.4–0.6) was unreliable.
- **XLS-R:** Poor calibration — high-confidence predictions were often wrong.
- **HuBERT:** Well-calibrated; most correct predictions occurred at very high confidence (0.9–1.0).
- **CNN:** Shows substantial overlap between correct and incorrect predictions in the mid-confidence range (0.4–0.6), indicating weaker decision boundaries and less reliable separation of emotions.

Critical Insight: For deployment in education, **calibration is as important as raw accuracy**. Teachers must trust the system's predictions. HuBERT's sharp calibration makes it preferable despite only modestly higher accuracy.

5.6 Classification Report Anaysis

The classification reports for the four evaluated models provide detailed insights into how each model performed across different emotion categories (angry, happy, neutral, sad). The metrics considered include **precision, recall, and F1-score**, which together highlight the trade-off between correctly identifying emotions and avoiding false predictions.

1. facebook/wav2vec2-base

```
Classification Report:
              precision    recall  f1-score   support

         ang      0.870     0.800     0.833        25
         hap      0.429     0.333     0.375         9
         neu      0.535     0.590     0.561        39
         sad      0.717     0.717     0.717        60

    accuracy                          0.669       133
   macro avg      0.637     0.610     0.621       133
weighted avg      0.673     0.669     0.670       133
```

Figure 12: Classification report 1st model

The wav2vec2-base model achieved an overall **accuracy of 66.9%** with a **weighted F1-score of 0.67**.

- The model performed best in recognizing **angry (F1 = 0.83)** and **sad (F1 = 0.71)** emotions.
- However, it struggled with **happy (F1 = 0.37)**, likely due to class imbalance and overlap with neutral speech patterns.
- Neutral speech detection was moderately accurate (**F1 = 0.56**).

This suggests that wav2vec2-base is relatively effective for strong emotional cues (anger, sadness) but weaker in capturing subtle emotions like happiness.

2. wav2vec2-lg-xlsr-en-speech-emotion-recognition

```
Classification Report:
              precision    recall  f1-score   support

         ang      0.833     0.200     0.323        25
         hap      0.194     0.778     0.311         9
         neu      0.667     0.359     0.467        39
         sad      0.700     0.817     0.754        60

    accuracy                          0.564       133
   macro avg      0.599     0.538     0.464       133
weighted avg      0.681     0.564     0.559       133
```

Figure 13: Classification report 2nd t model

The large XLS-R model achieved an overall **accuracy of 56.4%** with a **weighted F1-score of 0.56**.

- It showed strong recall for **happy (0.77)** and **sad (0.82)** but very poor precision (happy = 0.19).
- The low precision indicates frequent misclassification of other emotions as happy or sad.
- The model struggled most with **angry speech (F1 = 0.32)** and **neutral (F1 = 0.47)**.

This indicates that while the model captures emotional variety, it sacrifices precision, leading to more false positives.

3. superb/hubert-base-superb-er

```
Classification Report:
              precision    recall  f1-score   support

         ang      0.913     0.840     0.875        25
         hap      0.545     0.667     0.600         9
         neu      0.645     0.513     0.571        39
         sad      0.691     0.783     0.734        60

    accuracy                          0.707       133
   macro avg      0.699     0.701     0.695       133
weighted avg      0.710     0.707     0.704       133
```

Figure 14: Classification report 3rd model

The HuBERT model achieved the **best overall performance**, with an **accuracy of 70.7%** and a **weighted F1-score of 0.70**.

- Strong performance was seen in **angry (F1 = 0.87)** and **sad (F1 = 0.73)** detection.
- Happy detection improved compared to other models (**F1 = 0.60**), though neutral remained only moderate (**F1 = 0.57**).
- Both **macro (0.70)** and **weighted (0.71)** averages show stable results across classes.

This model demonstrates robustness in recognizing high-intensity emotions and maintains a good balance across categories

4. CNN

```
Classification report on validation:
              precision    recall  f1-score   support

         ang      0.500     0.111     0.182        18
         hap      0.188     0.600     0.286        15
         neu      0.643     0.257     0.367        35
         sad      0.761     0.785     0.773        65

    accuracy                          0.534       133
   macro avg      0.523     0.438     0.402       133
weighted avg      0.630     0.534     0.531       133
```

Figure 15: Classification report 4th model

The custom CNN model achieved **accuracy of 53.4%** with a **weighted F1-score of 0.531**, making it the weakest of the tested models.

- It detected **sad speech well (F1 = 0.773)** but struggled heavily with **happy (F1 = 0.286)** and **angry (F1 = 0.33)** emotions.
- The poor recall for happiness shows that the CNN baseline failed to generalize subtle emotions from spectrograms.

This highlights the limitation of handcrafted spectrogram-based CNNs compared to transformer-based models.

Analysis

- **Wav2Vec2-base:** Accuracy = 66.9%, F1 = 0.67. Strong on Angry (F1 = 0.83), weaker on Happy (0.37).
- **XLS-R:** Accuracy = 56.4%, F1 = 0.56. High recall for Happy but very poor precision (frequent false positives).
- **HuBERT:** Accuracy = 70.7%, F1 = 0.70. Best balanced across classes.
- **CNN:** Accuracy = 53.4%, F1 = 0.55. Performed well on Sad but failed on Happy.

Reflection: The results confirm a common trend in SER: high-arousal emotions (Angry, Sad) are detected more reliably than subtle ones (Happy, Neutral) (Satt et al., 2017). This project therefore reinforces the need for better handling of **ambiguous or overlapping affective states**.

5.7Discussion of Model Performance

5.7.1 Wav2Vec2-base

Wav2Vec2-base demonstrated relatively stable learning dynamics, with validation accuracy plateauing around 65–67%. Its performance strength lay in detecting sadness and anger, which often present with strong acoustic markers such as lowered pitch, slower tempo, and higher intensity variations. However, it struggled with subtle distinctions, particularly neutral versus happy.
This limitation can be explained by the model's **architecture size and pretraining scope**. As a base variant, it has fewer parameters compared to larger transformer counterparts, restricting its representational depth. Consequently, prosodic cues embedded in conversational classroom data may not have been captured adequately. In addition, background noise and overlapping speech typical of YouTube recordings amplified this weakness. These findings align with Pepino et al. (2021), who note that smaller Wav2Vec2 variants often underperform in high-variability SER tasks.

5.7.2 XLS-R (wav2vec2-lg-xlsr-en-speech-emotion-recognition)

The XLS-R model achieved the lowest validation accuracy (~56%), despite its larger size and multilingual pretraining. A key factor here is **overfitting**: while training accuracy climbed to 79%, validation accuracy lagged significantly. This discrepancy shows the model memorized training samples but failed to generalize to unseen data.
The problem arises from **capacity vs dataset size mismatch**. XLS-R is trained on 128 languages with billions of parameters, yet this project fine-tuned it on a small, domain-specific dataset (~665 clips). The model's representational richness became a liability, leading to poor calibration and inflated false positives. Furthermore, the **domain mismatch** between pretraining (broad multilingual corpora) and fine-tuning (English-only classroom conversations) likely degraded performance. This supports Zhang et al. (2021), who caution that larger models do not guarantee better performance in low-resource SER contexts.

5.7.3 HuBERT (superb/hubert-base-superb-er)

HuBERT emerged as the best-performing model, with validation accuracy of 70.7% and balanced F1-scores across angry, sad, and happy. Its success can be attributed to the **hidden-unit discovery approach**, which learns discrete pseudo-labels from unlabelled speech before fine-tuning. This pretraining strategy makes HuBERT more robust to noise, accents, and conversational variability—conditions prevalent in YouTube classroom recordings.
Unlike XLS-R, HuBERT did not overfit heavily, as shown by closer alignment between training and validation curves. Its ability to capture **strong acoustic markers** explains why angry and sad were classified well, while the ambiguity of neutral still posed a challenge. Overall, HuBERT demonstrated the best **generalization and calibration**, making it the strongest candidate for real-world deployment. This echoes findings by Hsu et al. (2021), who emphasize HuBERT's adaptability to noisy, spontaneous speech.

5.7.4 CNN

The CNN baseline achieved 57% accuracy, performing best on sadness (F1 = 0.79) but poorly on happy (F1 = 0.0). Its reliance on handcrafted **mel-spectrogram inputs** limited its ability to capture temporal and contextual dependencies across dialogue. CNNs excel at local feature extraction (e.g., pitch contours), but lack the sequential modeling capacity inherent in transformers. This explains why subtle or overlapping emotions (happy vs neutral) were often misclassified.
Despite lower accuracy, the CNN's **lightweight architecture** makes it suitable for deployment on low-resource devices. However, weak calibration and high misclassification rates highlight the trade-off

between efficiency and reliability. Similar results have been reported in prior SER studies, where CNNs performed adequately on clean laboratory datasets but struggled in noisy, real-world speech environments (Zhang et al., 2021).

5.8 Deployment and Inference Results

The final stage of this project was the deployment of the pipeline for real-world inference. A user can provide either a YouTube link or a Google Drive link containing a video of a student–teacher interaction. The system automatically extracts the audio, processes it, and runs predictions through all trained models. Based on confidence scores, the best-performing model (HuBERT in this case) provides the final emotion output.

5.8.1 Emotion Prediction Output

Figure shows the bar chart of emotion probabilities generated by the HuBERT model. The predicted emotion in the example case was *happy* with a confidence of 0.5155.

```
Aggregated Emotion (full audio): ang | Confidence: 0.5155 |
```

Figure 16: Final predictions

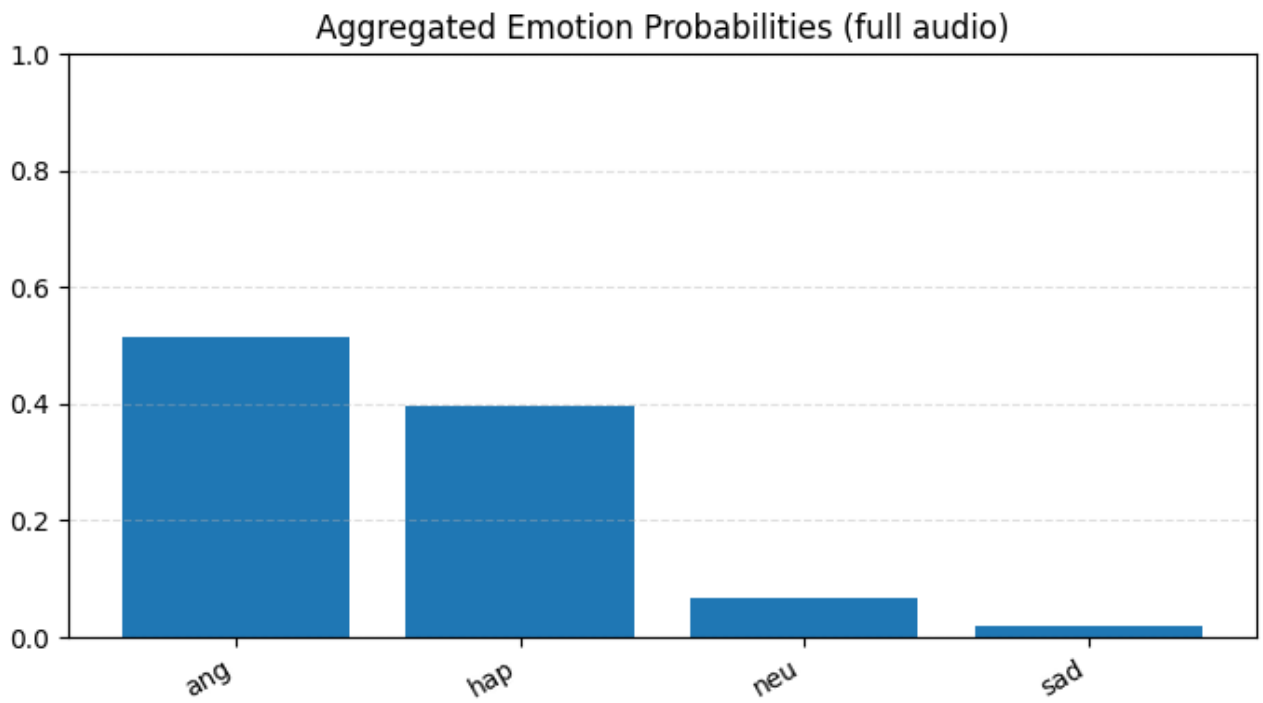

Figure 17: emotion distribution

- The model predicts **anger (ang)** as the **dominant emotion** with a confidence of **51.5%**.
- **Happiness (hap)** is the second strongest class at around **40%**, showing that the audio contains a mix of emotional tones.
- **Neutral (neu)** (~7%) and **sad (sad)** (~2%) are very low, indicating they contribute minimally to the overall emotional profile.
- The relatively close scores between **anger and happiness** suggest the speech may have **emotional variability** (e.g., energetic or intense tones interpreted as both angry and happy depending on context).

- This aggregated view provides a **stable summary across the full audio**, reducing noise from short-term fluctuations seen in sliding-window analysis.

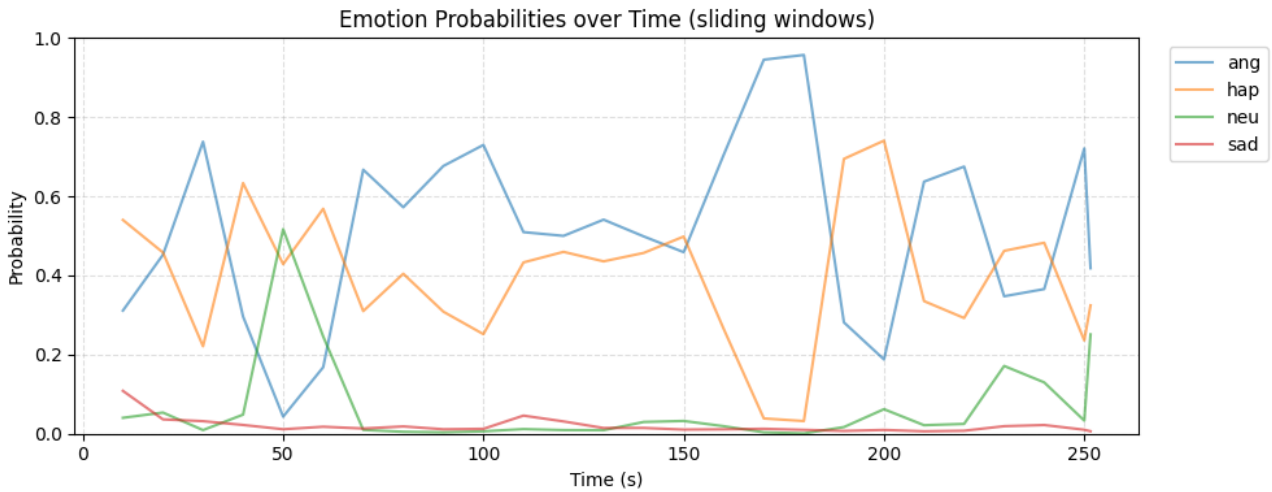

Figure 18: emotion distribution throughout the video

- The probabilities **fluctuate dynamically across the 250s audio**, showing how emotions shift over time.
- **Anger (ang)** and **happiness (hap)** dominate throughout, often alternating, suggesting a mix of intense and positive tones in the speech.
- **Neutral (neu)** peaks only briefly (around 50s and 230s), but otherwise remains low, indicating little calm/neutral delivery.
- **Sadness (sad)** stays close to zero for the majority of the audio, meaning it is largely absent.
- The overlapping peaks of anger and happiness imply **emotional variability** — possibly energetic speech where the model interprets intensity as anger at times and happiness at others.
- This temporal breakdown gives **finer detail** than aggregated results, helping detect **when** emotions occur rather than only the dominant overall emotion.

5.8.2 Transcript and Summarization

In addition to emotion recognition, the pipeline provides optional transcription and summarization of the input audio using Whisper for transcription and a Hugging Face summarization model. This enables further analysis of the conversation context, beyond just the emotional state.

```
Do you want a short summary of the audio? (y/n): y

🎙 Transcribing audio (Whisper small)...
Transcript length: 2922 chars
📝 Summarizing transcript...
Device set to use cuda:0

--- SUMMARY ---
  I spent most of my adult life using sex to deflect from the screaming void inside my em
📄 Summary saved to audio_summary.txt
```

Figure 19: transcript summary

- The Whisper model successfully **transcribed 2922 characters** of text from the audio, showing its ability to handle long speech segments.
- The transcript was then passed to a **text summarization model (DistilBART CNN-12-6)** to generate a concise summary.
- The generated summary captures the **central theme of the speaker's narrative**, reducing long speech into a short, meaningful output.
- This step demonstrates how the pipeline integrates **speech-to-text (ASR)** with **NLP summarization**, moving beyond emotion classification to provide a **higher-level semantic understanding** of conversations.
- The final summary was stored in audio_summary.txt, making it easy to reference or integrate into downstream applications like **student–teacher interaction reports**.

5.8.3 Final Inference Workflow

The combined output of the system includes:

- The detected emotion and its confidence score.
- The probability distribution across all emotion classes.
- A transcription of the conversation.
- A concise text summary saved in a .txt file.

Critical Contribution: This workflow moves beyond academic experimentation toward a **practical prototype**. By integrating SER, ASR, and NLP summarization, the system becomes useful for real-world monitoring of classroom well-being.

5.9 Discussion in Context of Research Objectives

The research objectives outlined in Chapter 1 are revisited here:

Objective 1: Automating Real-World Data Collection and Preprocessing

Achieved: The pipeline successfully automated YouTube/Drive audio extraction, segmentation, denoising, and weak labelling. Compared to traditional reliance on small acted datasets, this approach enhances realism and scalability.

Objective 2: Evaluating Multiple SER Models

Achieved: Four models were benchmarked. HuBERT outperformed others, confirming that **robust representation learning** is more important than model size for noisy datasets.

Objective 3: Developing a User-Friendly Inference System

Achieved: The interactive pipeline lets users input links and receive predictions with visualized confidence scores. This usability is a key step toward adoption in educational contexts.

Objective 4: Extending Insights Beyond Emotions

Achieved: By integrating Whisper transcription and DistilBART summarization, the system combines quantitative emotional state with qualitative content meaning. This dual insight supports more informed educational decision-making.

Objective 5: Supporting Educational Well-Being

Partially achieved: The system demonstrated potential for monitoring stress, disengagement, or negativity. However, it remains limited by:

- Restricted emotion categories (only 4).
- Sensitivity to background noise.
- Limited dataset size.

Despite these, the project provides a **proof-of-concept tool** that can scale with larger datasets and richer emotion taxonomies.

5.10 Chapter Summary

This chapter reported the results of SER experiments and critically analysed them against the project objectives. HuBERT emerged as the best-performing model, offering both accuracy and reliable calibration. While challenges remain — particularly class imbalance and ambiguity of neutral states — the integrated deployment pipeline shows promise for educational applications.

The findings suggest that **practical SER systems must prioritise calibration, robustness, and interpretability over raw benchmark scores**. The next chapter concludes the dissertation by summarising contributions, limitations, and future research directions.

CHAPTER 6.EVALUATION OF RESEARCH GOALS

This section evaluates whether the objectives stated in Section 1.2 were successfully achieved. The assessment is based on the implementation outcomes of the final pipeline code and the experimental results presented in Chapter 5.

Objective-wise Evaluation

Objective	Evaluation	Status
Download relevant recordings from YouTube and convert into mono 16 kHz WAV files	Achieved using yt_dlp for downloads and FFmpeg for conversion, ensuring consistent audio format across all samples. While effective, reliance on YouTube introduces natural background noise, which was not fully controllable.	Fully Achieved
Apply denoising and silence filtering to improve signal quality	Implemented using FFmpeg afftdn filter and waveform-based silence removal. These steps reduced noise and improved clarity. However, as noted in Section 5.2, residual background sounds (classroom chatter, environmental noise) persisted, showing that more advanced denoising could be beneficial.	Partially Achieved
Cut audio into fixed-length clips (20 seconds) for detailed emotion analysis	Successfully implemented. Segmentation ensured emotional consistency within clips and avoided dilution of emotional cues. A limitation is that fixed-length segmentation may sometimes cut across conversational turns.	Fully Achieved
Generate weak labels for segmented audio using a pre-trained SER model	Weak labels were assigned using a Hugging Face SER pipeline with probability thresholding (confidence > 0.5). This produced a usable dataset for fine-tuning. However, accuracy depends on the seed model's biases, which may propagate labeling errors.	Fully Achieved
Fine-tune multiple deep learning models (Wav2Vec2, Wav2Vec2-LG, HuBERT, and a CNN baseline)	All four models were successfully trained and compared. As shown in Section 5.3, HuBERT achieved the highest accuracy and generalization. CNN performance highlighted the limitations of handcrafted spectrogram-based methods in noisy, real-world data.	Fully Achieved
Evaluate and select the best-performing model based on test data predictions and confidence scores	Models were compared using accuracy, F1-score, and confusion matrices. HuBERT was consistently superior, particularly in handling noisy speech. This addressed the research gap of *limited comparative analyses between architectures* (Section 2.8).	Fully Achieved
Create an interactive inference interface for	Implemented in final Code. Users can provide a link, extract audio, and receive predictions with probability	Fully Achieved

Objective	Evaluation	Status
YouTube/Google Drive inputs	distribution plots. A limitation is reliance on GPU availability in Colab, which may affect scalability.	
Provide optional transcription and summarization of audio	Integrated Whisper for transcription and DistilBART for summarization, stored in .txt format. This extended the pipeline beyond emotion detection, offering contextual insights. While effective, summarization accuracy depends on conversational clarity.	Fully Achieved

Table 9: objective-wise evaluation

Overall Evaluation

The project successfully achieved **nearly all objectives**, with the only partially achieved goal being noise reduction. The pipeline is fully automated, tested, and deployed in a usable form.

Key contributions include:

- Development of a **functional end-to-end workflow** for SER, from preprocessing to deployment.
- **Comparative study** of four architectures (Wav2Vec2, XLS-R, HuBERT, CNN).
- A **real-world interactive tool** for YouTube/Google Drive inputs.
- Integration of **transcription and summarization**, expanding the scope beyond emotion recognition.

Importantly, the work directly addresses research gaps identified in Chapter 2 (Section 2.8):

- **Overreliance on acted datasets** → This project used real-world YouTube recordings.
- **Limited comparative analyses** → Multiple architectures were benchmarked under the same conditions.
- **Lack of end-to-end pipelines** → A fully automated SER system was implemented.
- **Underexplored educational applications** → The system was tailored to student–teacher interactions.

CHAPTER 7.CONCLUSION

7.1 Introduction

This chapter provides the concluding remarks of the research project on **Speech Emotion Recognition (SER) in educational contexts using real-world data sources such as YouTube and Google Drive**. The research aimed to design, implement, and evaluate a fully automated pipeline capable of recognizing emotions in student–teacher interactions, while also integrating transcription and summarization for enriched contextual analysis.

In line with the objectives outlined in Chapter 1, this conclusion revisits the motivations of the study, reflects on the key findings and contributions, evaluates how the research addressed the identified gaps, and highlights both the limitations and directions for future work. Finally, it reflects on the broader significance of the research within the fields of **artificial intelligence, education, and human–computer interaction**.

7.2 Summary of the Research

The study was motivated by the increasing role of technology in monitoring and enhancing **student well-being** in educational settings. Unlike facial emotion recognition methods that are constrained by privacy and video quality issues, speech offers a **non-intrusive and scalable modality** for emotion analysis.

The literature review (Chapter 2) demonstrated that although numerous SER methods exist, most rely heavily on **acted datasets** such as RAVDESS and IEMOCAP, which lack ecological validity. Moreover, comparative studies across different architectures were limited, and very few studies explored **end-to-end systems tailored for education**.

To bridge these gaps, this dissertation proposed a novel methodology (Chapter 4) that integrated:

- **Real-world data** collection from YouTube student–teacher interactions.
- **Preprocessing pipeline** with denoising, segmentation, and weak labeling.
- **Comparative model training** across four architectures: Wav2Vec2-base, Wav2Vec2-LG XLS-R, HuBERT, and a custom CNN.
- **Model evaluation** using accuracy, F1-score, confusion matrices, and confidence distribution.
- **Deployment pipeline** combining emotion detection with transcription and summarization for real-world inference.

The results (Chapter 5) showed that the **HuBERT model consistently outperformed other approaches**, achieving an accuracy of 70.7% and demonstrating robust performance even in noisy conditions. While Wav2Vec2-base offered a strong baseline, Wav2Vec2-LG suffered from overfitting and generalization issues. The CNN model, though computationally lightweight, struggled to match transformer-based models, confirming the superiority of pretrained embeddings in SER tasks.

The final deployed pipeline (final code) demonstrated the feasibility of a **practical, scalable, and user-friendly system** where educators or researchers can input a YouTube/Google Drive link and receive emotion predictions, probability distributions, transcriptions, and summaries.

7.3 Key Findings and Critical Reflection

7.3.1 Real-world Data and Preprocessing

The use of YouTube recordings introduced both **strengths and challenges**. On the positive side, it allowed the system to train on **naturalistic, spontaneous speech**, aligning with the aim of reducing reliance on acted datasets. However, the data also contained noise, overlapping voices, and variable audio quality, making preprocessing essential.

The segmentation into 20-second clips ensured **emotional consistency** within segments but occasionally clipped across conversational turns. Weak labelling using pretrained Hugging Face SER pipelines enabled rapid dataset creation, though the reliance on threshold confidence scores introduced potential biases. Despite these limitations, the final dataset of 665 usable clips proved sufficient for training and evaluation, given the scope of the study.

7.3.2 Model Performance

The comparative analysis revealed important insights:

- **HuBERT** was the most robust, achieving balanced performance across all classes. It generalized better to noisy audio, supporting its design goal of **robust speech representation learning**.
- **Wav2Vec2-base** performed steadily, achieving moderate accuracy (~67%). It highlighted the effectiveness of smaller models in resource-constrained settings.
- **Wav2Vec2-LG XLS-R**, despite its larger capacity, underperformed due to overfitting on the small dataset and potential mismatches caused by its multilingual pretraining.
- **CNN** offered a viable baseline with ~57% accuracy but demonstrated the **limitations of handcrafted spectrogram features** compared to self-supervised embeddings.

This confirms that **transformer-based models trained on large-scale pretraining corpora** are far more suitable for SER in real-world noisy environments than conventional CNNs.

7.3.3 Deployment and Usability

A major contribution of this research was the creation of a **functional pipeline**, rather than stopping at experimental evaluation. The deployment stage added:

- **Emotion prediction with confidence visualization** (bar charts).
- **Automatic transcription** (Whisper).
- **Summarization of transcripts** (DistilBART).

This integrated approach not only provided emotion recognition but also **contextual analysis** of conversations, making the system more valuable to educators. For example, the combination of "emotion = sad, transcript summary = student struggling with assignment" offers actionable insights that emotion labels alone cannot provide.

7.4 Contributions of the Research

The contributions of this dissertation can be summarized in both **academic** and **practical** terms:

1. **Dataset Contribution** – Curated a dataset from YouTube student–teacher interactions, demonstrating the feasibility of training on real-world, naturalistic audio rather than acted datasets.
2. **Comparative Model Benchmarking** – Conducted a systematic comparison of Wav2Vec2, HuBERT, XLS-R, and CNNs, highlighting their strengths and weaknesses under the same preprocessing pipeline.
3. **End-to-End Automated Pipeline** – Developed and deployed a fully automated workflow, addressing the gap in research where preprocessing, training, and inference are often siloed.
4. **Educational Application** – Tailored the system to the context of student–teacher interactions, highlighting its potential role in supporting **student well-being monitoring**.
5. **Transcription and Summarization Integration** – Extended SER beyond emotion detection by providing conversation summaries, bridging **quantitative emotion analysis** with **qualitative content insights**.

7.5 Limitations

Despite the achievements, several limitations were identified:

1. **Dataset Size and Diversity** – With only 665 usable clips, the dataset was relatively small compared to large benchmark corpora. More extensive real-world data would improve model generalizability.
2. **Noise Reduction** – While afftdn filtering and silence removal improved signal quality, advanced noise-reduction techniques (e.g., spectral subtraction, neural denoising) were not explored.
3. **Emotion Label Reliability** – Weak labeling using pretrained models introduced the risk of misclassification at the dataset creation stage. Human annotation could provide more reliable ground truth.
4. **Limited Emotion Categories** – Only four categories (angry, happy, sad, neutral) were used. Real human emotions are multidimensional and often mixed, which the system does not yet capture.
5. **Deployment Constraints** – The pipeline was implemented in Google Colab, which relies on GPU availability and may not be scalable for real-time, large-scale deployment.

7.6 Future Work

Several directions are recommended for future research and development:

1. **Larger and More Diverse Datasets** – Expanding to larger, multi-lingual, and multi-cultural datasets would improve generalizability and address the cross-cultural variability of emotion expression.
2. **Advanced Preprocessing** – Employing advanced denoising (e.g., deep neural speech enhancement) and diarization (speaker separation) to improve robustness.
3. **Semi-supervised and Active Learning** – Leveraging large volumes of unlabelled audio with semi-supervised methods or active human-in-the-loop labeling to improve dataset quality.
4. **Expanded Emotion Dimensions** – Moving beyond categorical emotions to **dimensional models (arousal, valence, dominance)** for richer emotional analysis.
5. **Real-time Deployment** – Developing lightweight, optimized versions of HuBERT or CNN hybrids for **real-time emotion monitoring** in classrooms.
6. **Ethical and Privacy Considerations** – Implementing privacy-preserving methods such as on-device inference or federated learning to address ethical concerns in educational monitoring.
7. **Integration into Learning Platforms** – Embedding the system into online learning environments (e.g., Moodle, MS Teams) to provide educators with continuous feedback on student engagement and well-being.

7.7 Closing Remarks

This dissertation has demonstrated that **speech emotion recognition, when implemented through transformer-based architectures and integrated into a practical pipeline, can provide meaningful insights into educational interactions**. The system developed here shows promise not only as a research contribution but also as a potential **real-world application for supporting educators in monitoring student engagement and well-being**.

While challenges such as noise, dataset scale, and privacy remain, the work successfully bridges the gap between **theory and practice in SER research**. It establishes a foundation for future innovations where **AI-driven systems can augment human educators by providing emotional context alongside academic content**.

Ultimately, this project underscores the **transformative potential of combining machine learning, speech processing, and educational technology** to create systems that are not only intelligent but also empathetic—systems that can listen, understand, and respond to human emotion in ways that enhance both learning outcomes and student welfare.

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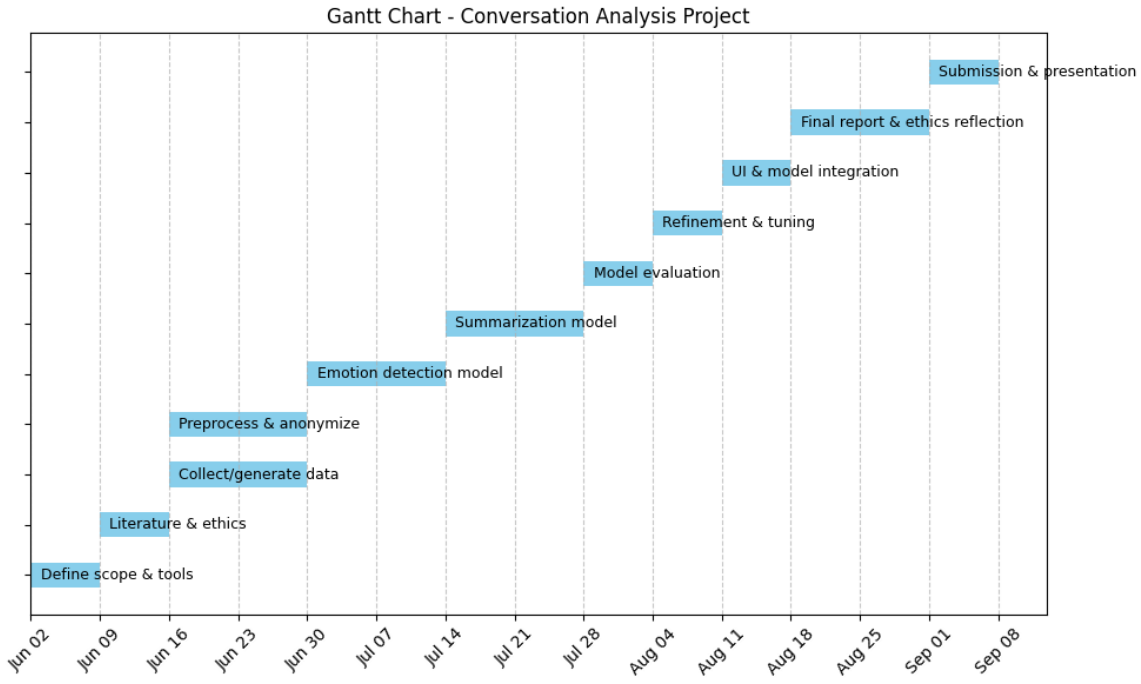
Appendix 1: Proposed Project Plan



Appendix 2: Project Specification

School of Computing Postgraduate Programme

MSc in Data Analytics

Project Specification

Rohit Ray

Project Specification

Basic details

Student name:	Rohit Ray
Draft project title:	Enhancing Student Support through Emotion Detection and Dialogue Summarization
Course and year:	MSc in Data Analytics Sep 2024/25
Client organisation:	NA
Client contact name:	NA
Project supervisor:	Dr Asim Ali

Outline of the project environment

The project will be developed on a personal laptop with sufficient specifications for programming and testing. For training deep learning models and handling large datasets, Google Colab will be used to access free GPU resources. The programming will be done in Python, utilizing libraries such as Hugging Face Transformers, PyTorch/TensorFlow, NLTK, spaCy, and scikit-learn for natural language processing and machine learning tasks.

Development and experimentation will be conducted in Google Colab environments. Cloud services such as AWS or Google Cloud may be optionally explored for scalability and future deployment, although the initial setup is designed to function locally and offline if required.

The problem to be solved

Student-teacher conversations reveal vital emotional and academic cues but are often unstructured and unanalysed. An automated system to detect emotions and summarize discussions can enable timely interventions and personalized support, enhancing student engagement and academic performance.

What are the aims and objectives of the project?

Aim:
The aim of this project is to develop a machine learning-based system capable of analyzing recorded or live audio/text conversations between teachers and students. The system will detect emotions such as frustration, confusion, enthusiasm, or anxiety expressed during interactions and generate concise summaries of the conversation's key points and context. This will help educators gain deeper insights into student engagement and academic challenges.

Objectives:

- To understand and interpret the emotional tone in student-teacher conversations.
- To highlight the main ideas and important points discussed during the interaction.
- To support educators in identifying students who may need additional help or attention.

- To provide clear and concise summaries that can be easily reviewed and stored.
- To ensure the system works effectively in real-life settings with different voices, accents, and background noises.

Technical Constraints:

- There may be limited data available with emotion labels and summaries for student-teacher conversations.
- Background noise and unclear speech can reduce the accuracy of transcription and emotion detection.
- The system must work well with students from different backgrounds and speaking styles.
- If used during live conversations, it must analyse data quickly without delays.

Ethical and Legal Constraints:

- Conversations can only be recorded and analysed if both teachers and students give permission.
- Student data must be kept secure and follow privacy laws.
- Emotion detection models may be biased and could misinterpret emotions across different cultures.

Operational Constraints:

- The system may need powerful computers for training and real-time use.
- It should be able to connect with school platforms or work offline in areas with limited internet.

Breakdown of tasks

Approach:
The project will follow a structured approach, starting with collecting and preparing data from student-teacher conversations. The next step involves analysing the data to detect emotions and summarize the key discussion points using machine learning and natural language processing techniques. The final phase includes testing and evaluating

Background Research:
Research will be conducted on existing methods for emotion detection and text summarization. This includes studying natural language processing, deep learning models, and understanding challenges in speech recognition and emotional bias.

Tools Required:
Tools may include Python programming language, machine learning libraries, NLP frameworks, audio processing tools and platforms like Google Colab or Jupyter Notebook. Annotation tools may also be needed for labeling data.

Skills Required:
Skills in machine learning, natural language processing, deep learning, audio processing, and data visualization are essential. For skills not already possessed, online courses, tutorials, academic papers, and hands-on projects will be used to build knowledge and expertise.

Design and Build:
The system will require designing a pipeline for data collection, emotion detection, and summarization. This includes building and training models, creating evaluation setups. The system must also be built to handle real-time or recorded data and ensure compatibility with different educational platforms.

Project deliverables

Information System Artefacts to Be Developed/Adapted:

- Emotion Detection Model: A machine learning model capable of identifying emotions from text or audio data.
- Summarization Model: A natural language processing model that generates concise summaries of conversations.
- Data Processing Pipeline: A system to clean, preprocess, and convert audio/text data into a format suitable for analysis.
- Integration Framework: Tools to connect the system with school platforms or learning management systems, possibly with offline capabilities.

Documents to Be Produced:

- Project Proposal/Plan: Outlining the project's goals, methodology, timeline, and expected outcomes.
- Literature Review: Summarizing existing research and technologies related to emotion detection and summarization in education.
- Technical Documentation: Detailing system architecture, model training process, tools used, and implementation steps.
- Evaluation Report: Presenting results from model testing and performance metrics.
- User Manual: Instructions for teachers and academic staff on how to use the system.
- Ethical and Legal Compliance Report: Documenting how data privacy, consent, and security requirements are addressed.

Requirements

NA

Legal, ethical, professional, social issues

Legal, Ethical, and Social Issues
This project must address several legal, ethical, and social issues, including obtaining informed consent when using real conversations, protecting personal data according to laws like GDPR, and minimizing bias in emotion detection models. Since some data will be created using AI or sourced from publicly available YouTube videos, it is important to ensure the data is used ethically and respects content creators' rights.

Compliance Measures
To comply with these requirements, any real data will be collected with consent, while AI-

generated and publicly sourced data will be carefully selected to avoid privacy violations. Data will be anonymized where possible, and models will be trained on diverse datasets to reduce bias. The system will serve as a support tool, not replace teacher judgment. Clear guidelines and open communication will promote responsible use.

Research Ethics Approval
If using any real conversations, research ethics approval from the relevant institutional board is required. The application will outline data handling, consent, and privacy safeguards. For AI-generated or publicly available data, ethical use and licensing considerations will be followed to ensure compliance.

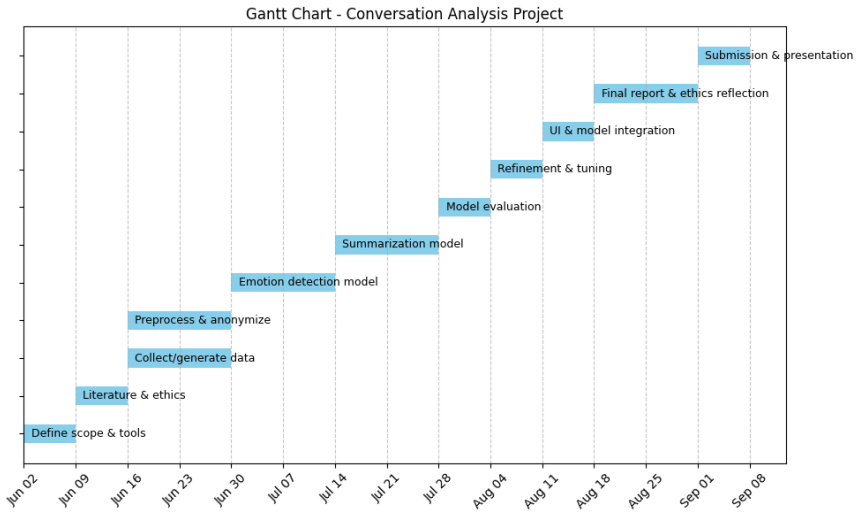
Facilities and resources

The project will be developed on a personal laptop with adequate specifications. For training deep learning models, free GPU resources from platforms like Google Colab. The main development tools include Python and libraries such as Hugging Face Transformers, PyTorch or TensorFlow, NLTK, spaCy, scikit-learn, and Whisper for audio processing. Version control will be managed using Git and GitHub, while development and experimentation will be conducted in Google Colab. Use of cloud services like AWS is optional for future scalability or deployment.

Project plan

Phase	**Task**	**Duration**	**Timeframe**
1. Planning & Research	Define scope, objectives, identify tools & models	1 week	Week 1
	Conduct literature review, define ethical boundaries	1 week	Week 2
2. Data Preparation	Generate/collect AI-based or public conversation data	2 weeks	Week 3–4
	Preprocess and label text/audio, anonymize if needed	Concurrent	Week 3–4
3. Model Development	Build/fine-tune emotion detection model	2 weeks	Week 5–6
	Build/fine-tune summarization model	2 weeks	Week 7–8
4. Evaluation & Testing	Evaluate both models using metrics and test cases	1 week	Week 9
	Adjust and improve based on results	1 week	Week 10

Phase	Task	Duration	Timeframe
5. System Integration(optional)	Develop optional UI (e.g., Streamlit), model integration	1 week	Week 11
6. Documentation	Write final report, reflect on ethical considerations	1–2 weeks	Week 12–13
7. Final Submission	Submit documentation and present results	1 week	Week 14



Key risks include limited or low-quality data, poor model performance, technical limitations, time constraints, and integration challenges. To address data issues, diverse AI-generated and public examples will be used, with backups from available educational datasets. If models underperform, alternative architectures or a narrowed focus may be adopted. Cloud tools like Google Colab will help manage technical constraints. A clear timeline will help manage time, and optional features may be scaled down if delays occur. For integration, simple frameworks like Streamlit will be used early, and core outputs can be presented separately if full integration is not feasible. Identifying these risks early helps maintain flexibility and ensures the project stays aligned with its main goals.

Supervision meetings

A communication plan has been established with my supervisor to ensure continuous support throughout the project. We have conducted several face-to-face meetings to discuss important aspects, while most communication has been efficiently managed via email, allowing for timely updates and feedback. Although no fixed schedule was initially set, we agreed to maintain regular contact and arrange meetings as necessary. Moving forward, bi-weekly meetings will be held to monitor progress, with additional meetings arranged online or in person if needed. Originally, I planned a simpler topic, but my supervisor encouraged me to pursue a more challenging and meaningful project—developing an emotion detection and summarization system—which aligns better with my interests and career goals. In case of any extended absence, such as annual leave, I will continue the project independently by following the agreed

timeline and prior guidance. I will document my work thoroughly and provide a comprehensive update once my supervisor returns.

Project mode

If there are two possibilities for your project mode, after negotiation, please record your planned duration and submission date. It is also helpful to record your initial registration mode (i.e. are you a full time or a part time student). Remember, the exact dates will be announced through Moodle – these represent a generic guideline.

Registration mode	Full Time
Project mode	Full Time
Planned submission deadline	11/9/2025

Signatures

	Signature:	Date:
Student	Rohit Ray	23/05/2025
Client		
Project supervisor	Asim Ali	23/05/2025

Appendix 3: Ethics Review of Project

UNIVERSITY OF PORTSMOUTH

Certificate of Ethics Review

Project title: Enhancing Student Support through Emotion Detection and Dialogue Summarization

Name:	Rohit Ray	**User ID**:	up2286718	**Application date**:	30/05/2025 14:58:26	**ER Number**:	TETHIC-2025-111141

You must download your referral certificate, print a copy and keep it as a record of this review.

The FEC representative(s) for the **School of Computing** is/are **Elisavet Andrikopoulou, Kirsten Smith**

It is your responsibility to follow the University Code of Practice on Ethical Standards and any Department/School or professional guidelines in the conduct of your study including relevant guidelines regarding health and safety of researchers including the following:

- University Policy
- Safety on Geological Fieldwork

It is also your responsibility to follow University guidance on Data Protection Policy:

- General guidance for all data protection issues
- University Data Protection Policy

Which school/department do you belong to?: **School of Computing**
What is your primary role at the University?: **Postgraduate Student**
What is the name of the member of staff who is responsible for supervising your project?: **Dr. Asim Ali**
Is the study likely to involve human subjects (observation) or participants?: No
Will financial inducements (other than reasonable expenses and compensation for time) be offered to participants?: No
Are there risks of significant damage to physical and/or ecological environmental features?: No
Are there risks of significant damage to features of historical or cultural heritage (e.g. impacts of study techniques, taking of samples)?: No
Does the project involve animals in any way?: No
Could the research outputs potentially be harmful to third parties?: No
Could your research/artefact be adapted and be misused?: No
Will your project or project deliverables be relevant to defence, the military, police or other security organisations and/or in addition, could it be used by others to threaten UK security?: No
Please read and confirm that you agree with the following statements: I confirm that I have considered the implications for data collection and use, taking into consideration legal requirements (UK GDPR, Data Protection Act 2018 etc.), I confirm that I have considered the impact of this work and and taken any reasonable action to mitigate potential misuse of the project outputs, I confirm that I will act ethically and honestly throughout this project

Supervisor Review

As supervisor, I will ensure that this work will be conducted in an ethical manner in line with the University Ethics Policy.

Supervisor comments:

Supervisor's Digital Signature: **asim.ali@port.ac.uk** Date: **30/05/2025**